数据库查询：过程与截图

NHLI70037 ICA resit 2026 · Bob · SCN1A 变异（题目写法 c.3993C>T，p.Asp1331Glu）
查询时间：2026-11-04 23:30–23:57 UTC（每张图的来源栏有精确访问时间）　|　文档生成：2026-11-05
浏览器：云端容器内 Chromium（Playwright 驱动），不是你的 Chrome（本会话没有 Claude in Chrome）。黄色高亮、红色圆角框和顶部来源栏是我加的，页面内容未改动。
NCBI 的 nuccore、ClinVar、PMC 网页对该浏览器弹出验证码，PubMed 返回 403，按规则未点击、未绕过；相关查询改用 NCBI E-utilities 官方接口在浏览器中显示，或用等效公开数据库（gnomAD、VariantValidator、Mutalyzer、UniProt）代替。
2026-11-05 修订版：自查后重拍了标签遮挡页面内容或标注错误的截图，新增 D30–D35，修正了几处没有依据的表述，详见文末“修订记录”。

0. 结论汇总

#	问题	结果	依据
1	题目写法 c.3993C>T (p.Asp1331Glu) 是否成立	不成立。c. 与 p. 互相矛盾（第 3 位 C>T 只能把 GAC 变成 GAT，仍是 Asp）；MANE 转录本 c.3993 参考碱基为 A，第 1331 位为 Glu	D02–D07
2	参考转录本	NM_001165963.4 = MANE Select（= ENST00000674923.1），编码 NP_001159435.1（2009 aa；与 UniProt P35498 序列一致，脚本比对）	D01、D06
3	与参考序列相容的候选变异	可证明：在 GAA 密码子上能产生 Asp 的单碱基替换只有 c.3993A>T 和 c.3993A>C，均为 p.(Glu1331Asp)；GRCh38 chr2:166009728 T>A / T>G（GRCh37 166866238）。推断：出题人想写的可能是其中之一，出错原因无法判断	D02、D06
4	gnomAD v4.1.2	该位点没有任何变异（已勾选 Indels 和 Filtered variants）；覆盖良好；相邻位点 AN ≈ 1,605,000	D08–D11
5	ClinVar	第 1331 位无记录；邻近 1328、1329、1333 位有 P/LP 错义	D12、D13
6	旁系同源对应位点	SCN2A p.Glu1321Gln = LP（1 个提交者）；SCN8A p.Glu1311Val = VUS；SCN3A 无记录	D14、D29
7	基因层面	Dravet、GEFS+、DEE 均 Definitive；HI score 3；错义 Z = 8.78；pLoF o/e = 0.07（pLI = 1）	D01、D15、D16
8	蛋白位置	E1331 位于 UniProt 标注的 “S4 of repeat III” 跨膜螺旋（1314–1332）末端，其后 1333–1345 为胞质侧（推断为 S4–S5 连接段）	D17
9	计算预测（p.Glu1331Asp）	REVEL 0.613：按 Pejaver 2022 表 2（经 PMC 全文接口核对，未截图），PP3 需 ≥0.644、BP4 需 ≤0.290，两者都不满足。AlphaMissense 0.8497；CADD 23–26（版本不同）；SpliceAI 全部 0.00	D07、D18–D21
10	保守性（手动比对）	人类全部 9 个 Nav α 亚基（SCN1A 及 8 个旁系同源）以及所查哺乳动物、鸡、爪蟾、斑马鱼 scn1lab 均为 E；斑马鱼 scn1laa 为 D（5 个 TrEMBL 条目，同一基因），果蝇 para 为 Q	表 A1
11	ClinGen VCEP SCN1A 细则 v2.0	PM1 适用（PER #10 含 1331）；PM2 仅 Supporting；PP2、PP4 不适用；PP3/BP4 用 REVEL；PS2/PM6 按表型计分；合并规则要用附件 PDF（未下载）	D22–D35

A. 命名与参考序列核对

推导（不依赖任何数据库）：密码子 n 占 c.(3n−2)～c.3n；3993 = 3 × 1331，所以 c.3993 是第 1331 号密码子的第 3 位。Asp 的密码子是 GAT/GAC，Glu 是 GAA/GAG（图 D03）。第 3 位 C>T 只能把 GAC 变成

GAT（仍是 Asp，同义）；Asp→Glu 必须是第 3 位变成 A 或 G。因此 “c.3993C>T (p.Asp1331Glu)” 在任何转录本上都不可能同时成立。

A1　参考转录本是 MANE Select

gnomAD 基因页列出 MANE Select = ENST00000674923.1 / NM_001165963.4（D01）；VariantValidator 结果也标注 Transcript Flag: MANE Select（D06）。【需查】NCBI RefSeq 页面（KEYWORDS “MANE Select”）因验证码未能截图；我用 E-utilities 下载的 GenBank 记录核对过：KEYWORDS = “RefSeq; MANE Select.”，CDS 479..6508，转录本第 4469–4471 位 = gaa（c.3991–3993）。需要的话请在你的 Chrome 打开 https://www.ncbi.nlm.nih.gov/nuccore/NM_001165963.4 补截。

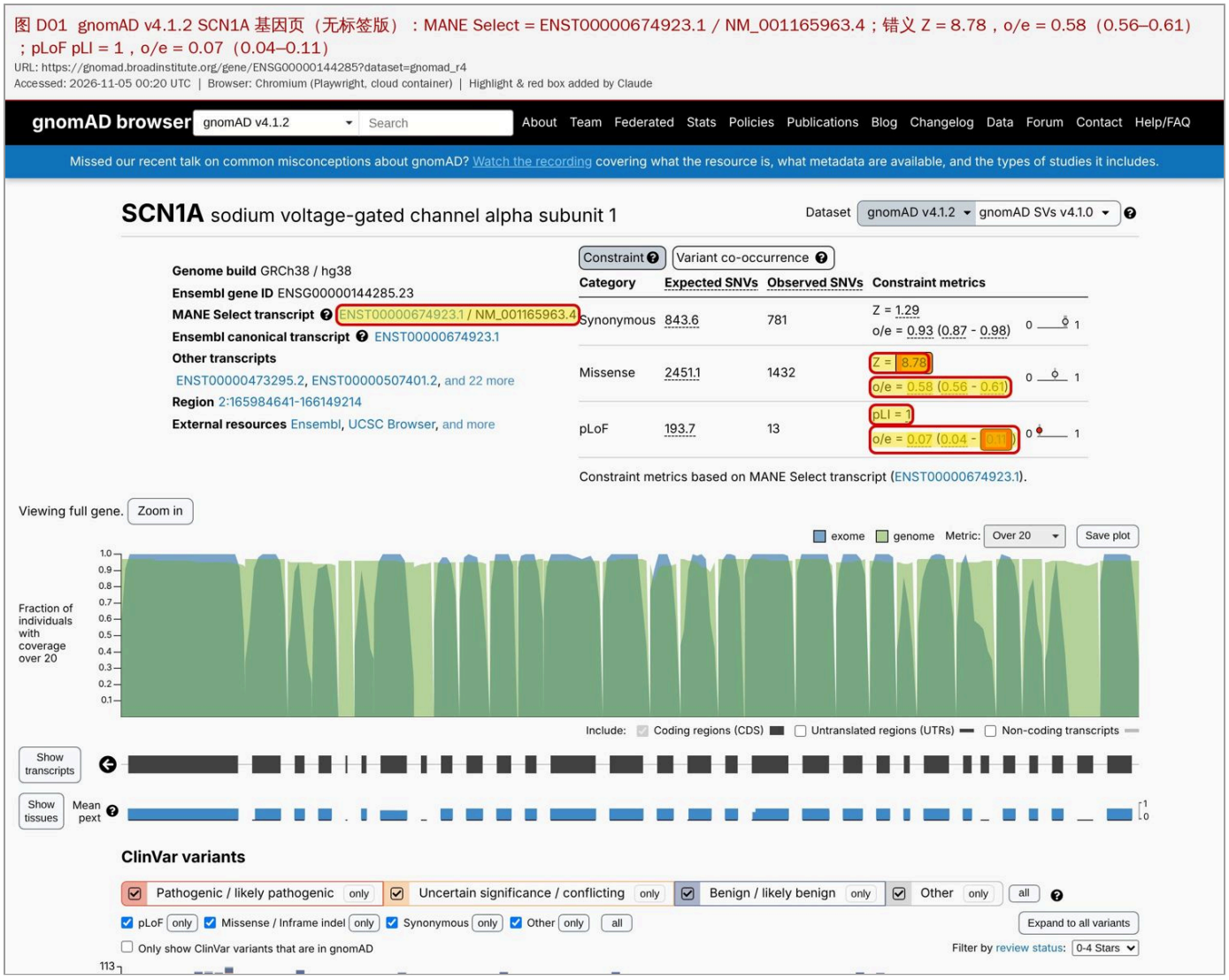


图 D01　gnomAD v4.1.2 SCN1A 基因页（无标签版）：MANE Select = ENST00000674923.1 / NM_001165963.4；错义 Z = 8.78，o/e = 0.58（0.56–0.61）；pLoF pLI = 1，o/e = 0.07（0.04–0.11）

文件：screenshots/D01_gnomAD_MANE与基因约束.jpg　|　来源：https://gnomad.broadinstitute.org/gene/ENSG00000144285?dataset=gnomad_r4　|　访问：2026-11-05 00:20 UTC

A2　遗传密码

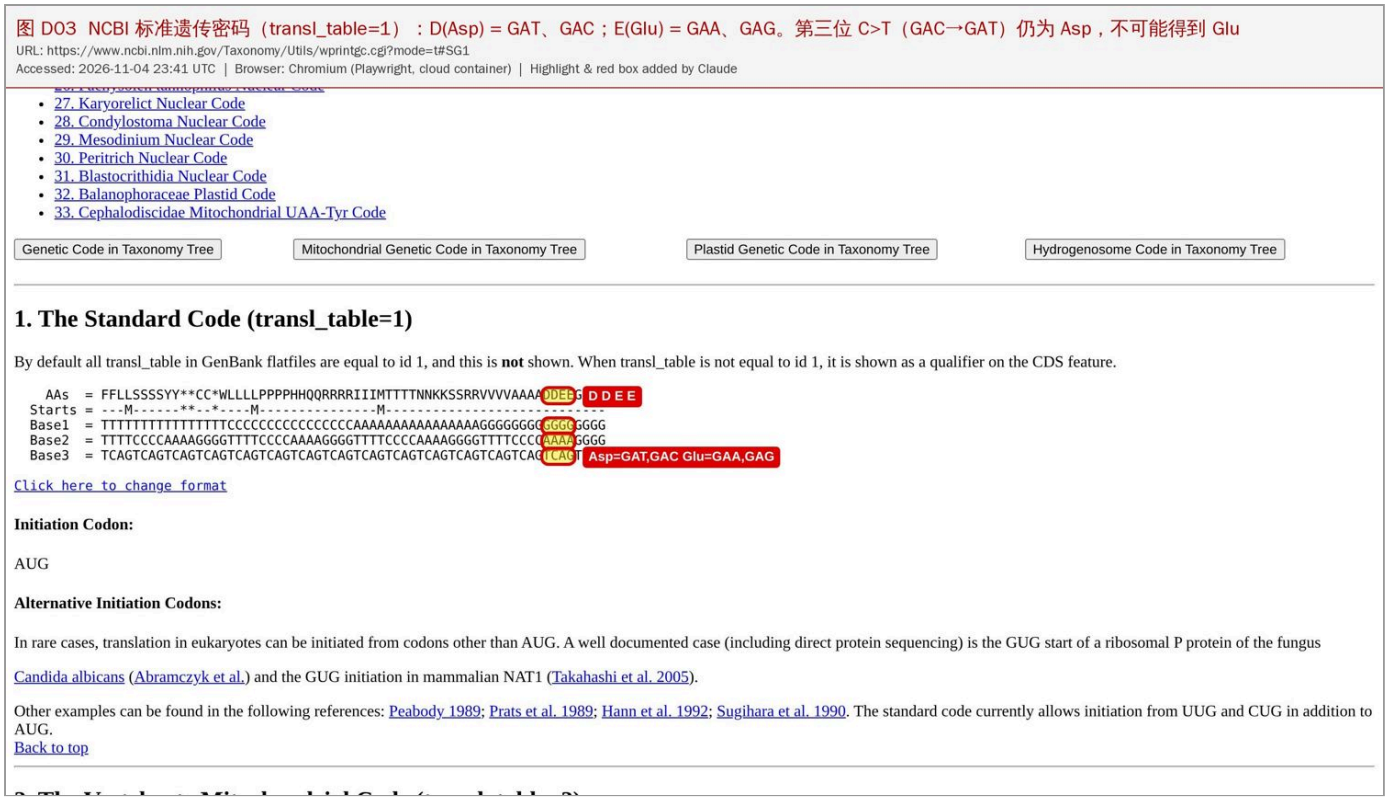


图 D03　NCBI 标准遗传密码（transl_table=1）：D(Asp) = GAT、GAC；E(Glu) = GAA、GAG。第三位 C>T（GAC→GAT）仍为 Asp，不可能得到 Glu

文件：screenshots/D03_NCBI_标准遗传密码表_Asp_Glu.jpg ｜ 来源：https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi?mode=t#SG1 ｜ 访问：2026-11-04 23:41 UTC

A3　题目写法校验：两个独立工具都报参考碱基不符

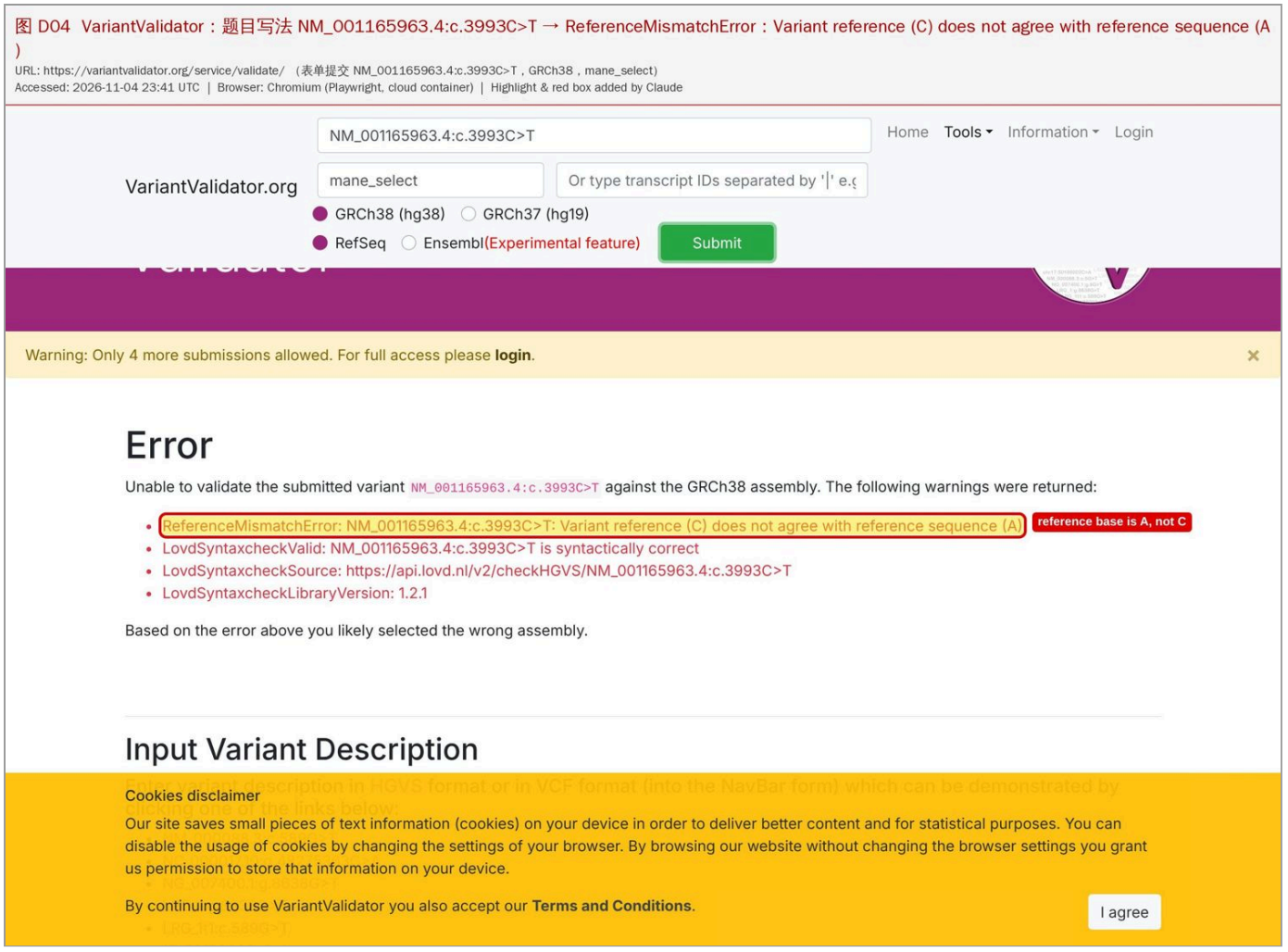


图 D04　VariantValidator：题目写法 NM_001165963.4:c.3993C>T → ReferenceMismatchError：Variant reference (C) does not agree with reference sequence (A)

文件：screenshots/D04_VariantValidator_c3993CT_参考碱基不符.jpg　|　来源：https://variantvalidator.org/service/validate/ （表单提交 NM_001165963.4:c.3993C>T，GRCh38，mane_select）　|　访问：2026-11-04 23:41 UTC



图 D05　Mutalyzer 3（独立第二个工具）：c.3993C>T → "`C` was not found in the reference sequence; `A` was found instead."

文件：screenshots/D05_Mutalyzer_c3993CT_参考碱基不符.jpg ｜ 来源：https://mutalyzer.nl/normalizer/NM_001165963.4:c.3993C%3ET ｜ 访问：2026-11-04 23:42 UTC

A4　推断写法校验：c.3993A>T 成立，给出 p.(Glu1331Asp) 和基因组坐标

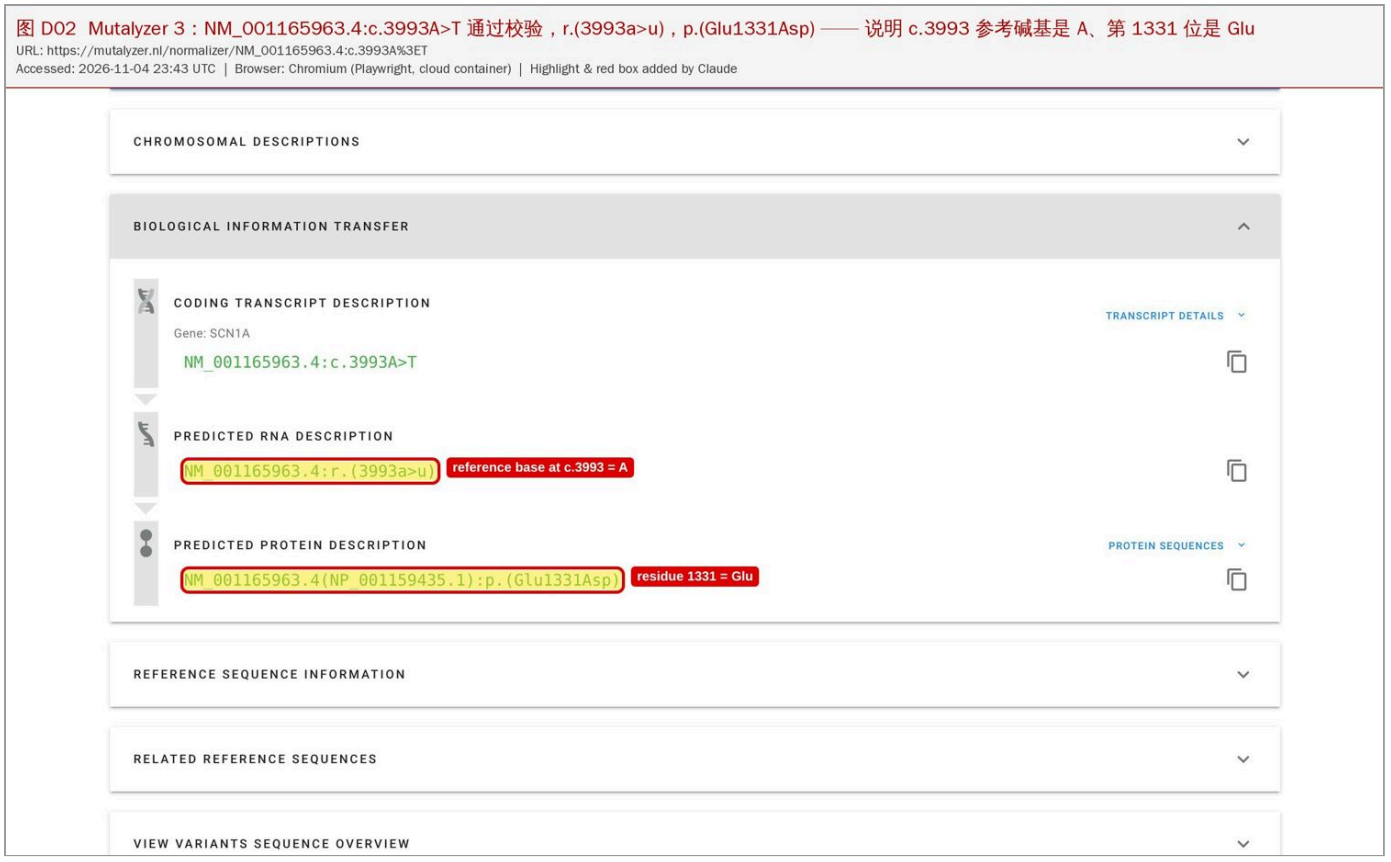


图 D02　Mutalyzer 3：NM_001165963.4:c.3993A>T 通过校验，r.(3993a>u)，p.(Glu1331Asp) —— 说明 c.3993 参考碱基是 A、第 1331 位是 Glu

文件：screenshots/D02_Mutalyzer_c3993AT_参考碱基A_Glu1331.jpg ｜ 来源：https://mutalyzer.nl/normalizer/NM_001165963.4:c.3993A%3ET ｜ 访问：2026-11-04 23:43 UTC

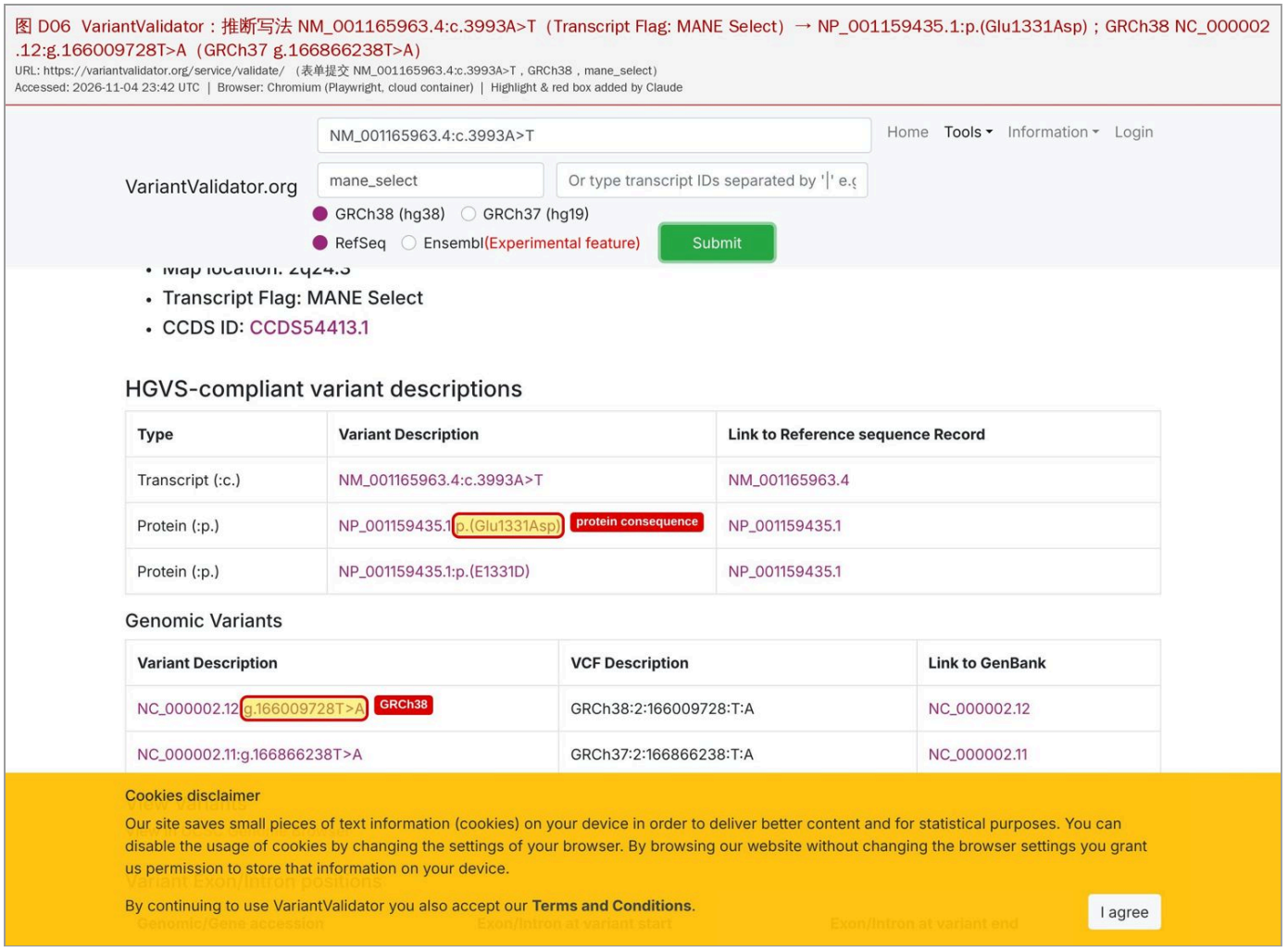


图 D06　VariantValidator：推断写法 NM_001165963.4:c.3993A>T（Transcript Flag: MANE Select）→ NP_001159435.1:p.(Glu1331Asp)；GRCh38 NC_000002.12:g.166009728T>A（GRCh37 g.166866238T>A）

文件：screenshots/D06_VariantValidator_c3993AT_Glu1331Asp_GRCh38.jpg ｜ 来源：https://variantvalidator.org/service/validate/ （表单提交 NM_001165963.4:c.3993A>T，GRCh38，mane_select） ｜ 访问：2026-11-04 23:42 UTC

A5　dbNSFP 注释：参考密码子 GAA、第 3 位、第 1331 位

图 D07　MyVariant.info（dbNSFP，GRCh37 chr2:166866238 T>A）。高亮按出现顺序：CADD phred 25.9；alt = D；codonpos 3；pos 1331；ref = E；refcodon GAA；REVEL score 0.613

URL: https://myvariant.info/v1/variant/chr2:g.166866238T%3EA?assembly=hg19&fields=dbnsfp.revel,dbnsfp.alphamissense,cadd.phred,dbnsfp.aa
Accessed: 2026-11-05 00:20 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

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```
{"_id":"chr2:g.166866238T>A","_version":1,"cadd":{"_license":"http://bit.ly/2TIuab9","phred":25.9},"dbnsfp":{"_lice
nse":"http://bit.ly/2VLnQBz","aa":{"alt":"D","codon_degeneracy":[2,2,2,2,2,2,2,2,2,2],"codonpos":[3,3,3,3,3,3,3,3,
3,3],"pos":[1319,1331,1320,1331,1320,1320,1331,1320,1320,1303],"ref":"E","refcodon":["GAA","GAA","GAA","GAA","GA
A","GAA","GAA","GAA","GAA","GAA"]},"alphamissense":{"pred":["P","P","P","P","P","P","P","P","P","P"],"rankscore":0.88414,"scor
e":[0.9583,0.9506,0.9628,0.9536,0.8497,0.7963,0.9584]},"revel":{"rankscore":0.84879,"score":[0.613,0.613,0.613,0.613,0.61
3]}}}
```


图 D07　MyVariant.info（dbNSFP，GRCh37 chr2:166866238 T>A）。高亮按出现顺序：CADD phred 25.9；alt = D；codonpos 3；pos 1331；ref = E；refcodon GAA；REVEL score 0.613

文件：screenshots/D07_dbNSFP_参考密码子GAA_REVEL.jpg ｜ 来源：https://myvariant.info/v1/variant/chr2:g.166866238T%3EA?assembly=hg19&fields=dbnsfp.revel,dbnsfp.alphamissense,cadd.phred,dbnsfp.aa ｜ 访问：2026-11-05 00:20 UTC

补充（脚本核对，未截图）：我把 NCBI 上 5 个 SCN1A RefSeq 转录本逐一翻译，c.3993 在全部转录本上都是 A；第 1331 位分别为 E（NM_001165963.4、NM_001202435.3、NM_001353948.2）、G（NM_006920.6）、L（NM_001165964.3），没有一个是 Asp。

B. 人群数据（gnomAD v4.1.2）

1. 打开区域 2-166009698-166009758（c.3993 前后各 30 bp），勾选 SNVs、Indels、Filtered variants、Exomes、Genomes（D08）。
2. 变异表按位置排序：2-166009725（c.3996）之后直接是 2-166009733（c.3988），2-166009728 没有任何变异。相邻位点 AN ≈ 1,605,000 个等位基因，远超 VCEP 要求的 10,000。
3. 覆盖度图（D09）：61 根柱中第 31 根 = chr2:166009728。截图只能目测柱高，精确数值来自 gnomAD API（未截图）：外显子 mean 29.3×、98% 个体 ≥20×；基因组 mean 30.6×、94% ≥20×。
4. 直接打开两个可能变异的页面，均为 “Variant not found”（D10、D11）。gnomAD API 另外确认 2-166009728-T-C 也不存在（该位点三种替换都没有）。



图 D08　gnomAD v4.1.2 区域 2-166009698-166009758（无标签版；已勾选 Indels、Filtered variants）：2-166009725（p.Gly1332Gly）之后直接是 2-166009733（p.Phe1330Leu），中间没有 2-166009728；两行的 Allele Number 分别为 1,605,234 和 1,605,814

文件：screenshots/D08_gnomAD_区域变异表_166009728无变异.jpg ｜ 来源：https://gnomad.broadinstitute.org/region/2-166009698-166009758?dataset=gnomad_r4 （已勾选 SNVs、Indels、Filtered variants、Exomes、Genomes） ｜ 访问：2026-11-05 00:20 UTC

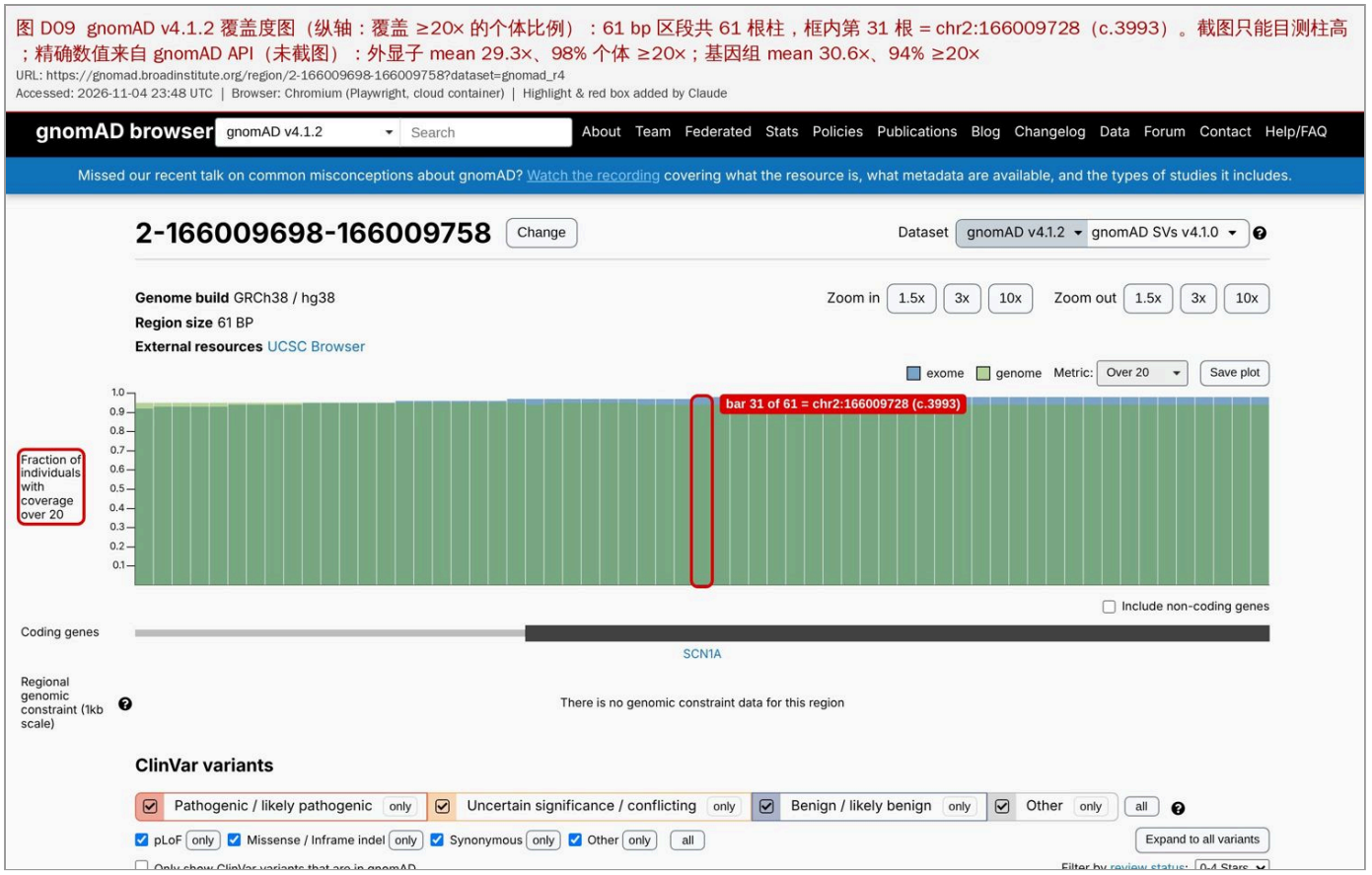


图 D09　gnomAD v4.1.2 覆盖度图（纵轴：覆盖 ≥20× 的个体比例）：61 bp 区段共 61 根柱，框内第 31 根 = chr2:166009728（c.3993）。截图只能目测柱高；精确数值来自 gnomAD API（未截图）：外显子 mean 29.3×、98% 个体 ≥20×；基因组 mean 30.6×、94% ≥20×

文件：screenshots/D09_gnomAD_覆盖度_166009728.jpg　|　来源：https://gnomad.broadinstitute.org/region/2-166009698-166009758?dataset=gnomad_r4　|　访问：2026-11-04 23:48 UTC

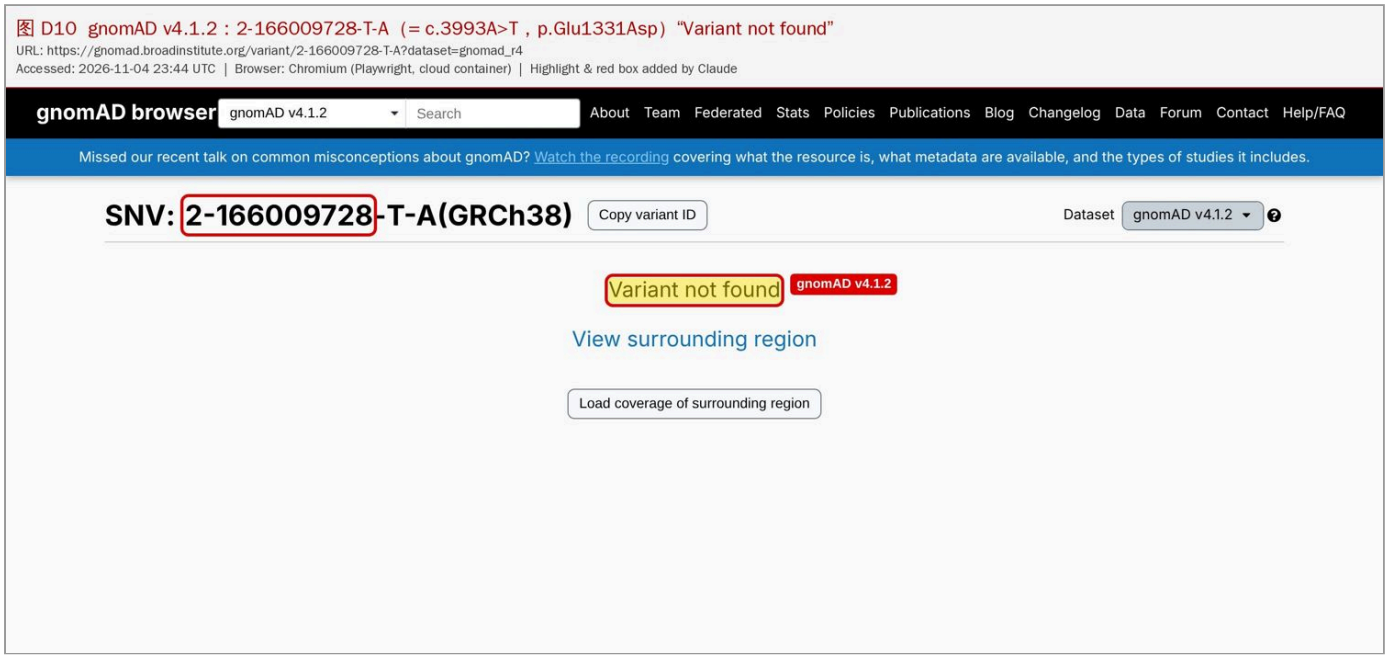


图 D10　gnomAD v4.1.2：2-166009728-T-A（= c.3993A>T，p.Glu1331Asp）“Variant not found”

文件：screenshots/D10_gnomAD_2-166009728-T-A_未找到.jpg　|　来源：https://gnomad.broadinstitute.org/variant/2-166009728-T-A?dataset=gnomad_r4　|　访问：2026-11-04 23:44 UTC

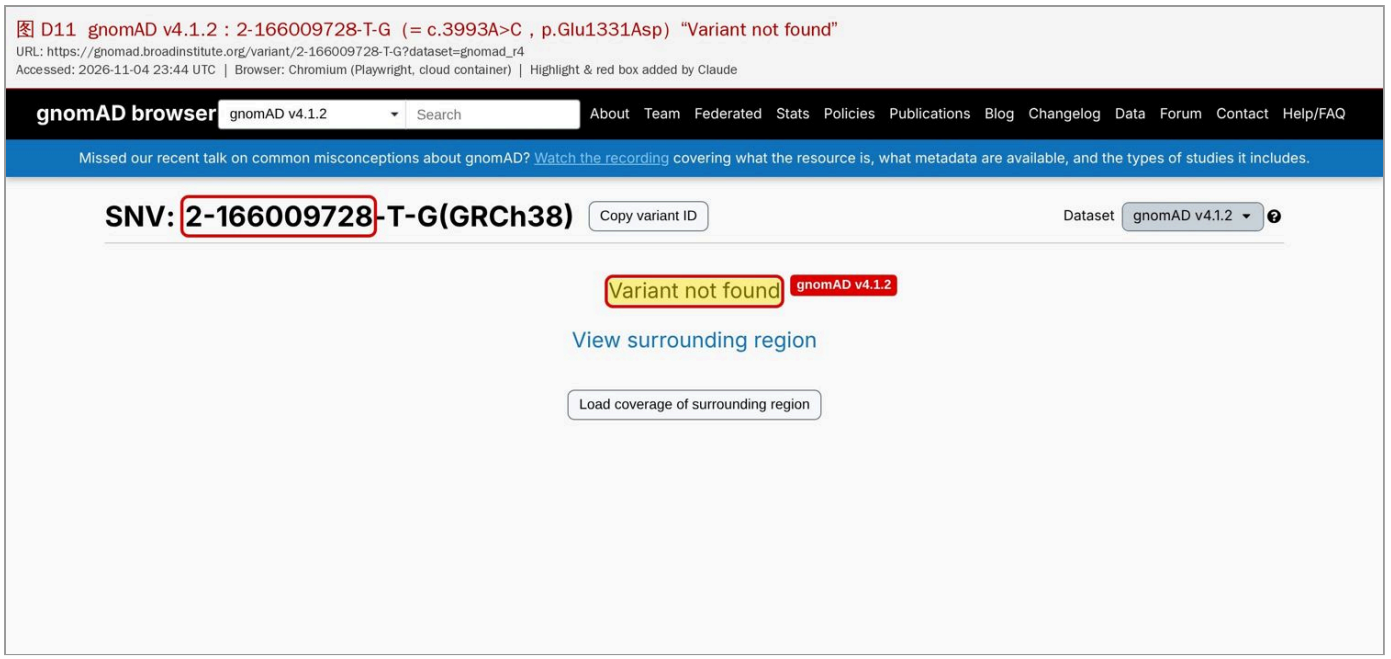


图 D11　gnomAD v4.1.2：2-166009728-T-G（= c.3993A>C，p.Glu1331Asp）“Variant not found”

文件：screenshots/D11_gnomAD_2-166009728-T-G_未找到.jpg ｜ 来源：https://gnomad.broadinstitute.org/variant/2-166009728-T-G?dataset=gnomad_r4 ｜ 访问：2026-11-04 23:44 UTC

C. 临床数据库

ClinVar 网页对容器浏览器弹验证码（未点），改用 NCBI E-utilities 在浏览器中显示检索结果。另用脚本下载了 SCN1A 全部 5,556 条 ClinVar 记录的摘要，按标题筛选第 1331 位：0 条；邻近 1328–1333 位的记录见 D13。



图 D12　ClinVar（NCBI E-utilities 检索，无标签版；ClinVar 网页对容器浏览器弹验证码，未点）：检索式 2[chr] AND 166009728:166009730[chrpos38]（第 1331 号密码子）→ count = 0

文件：screenshots/D12_ClinVar_第1331位无记录.jpg ｜ 来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=clinvar&term=2%5Bchr%5D+AND+166009728%3A166009730%5Bchrpos38%5D&retmode=json ｜ 访问：2026-11-05 00:20 UTC

图 D13 ClinVar（E-utilities esummary，无标签版）：高亮依次为 p.Ser1328Pro / Pathogenic/Likely pathogenic；p.Arg1329Pro / Pathogenic；p.Arg1329Leu / Pathogenic；p.Met1333Ile / Likely pathogenic

URL: https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2203183,2169505,1916226,2430161&retmode=json
Accessed: 2026-11-05 00:20 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

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{"header":{"type":"esummary","version":"0.3"},"result":{"uids":["2203183","2169505","1916226","2430161"],"2203183":{"uid":"2203183","obj_type":"single nucleotide variant","accession":"VCV002203183","accession_version":"VCV002203183.5","title":"NM_001165963.4(SCN1A):c.3982T>C (p.Ser1328Pro)","variation_set":[{"measure_id":"1925411","variation_xrefs":[{"db_source":"ClinGen","db_id":"CA349052944"},{"db_source":"dbSNP","db_id":"2468493029"}],"variation_name":"NM_001165963.4(SCN1A):c.3982T>C (p.Ser1328Pro)","cdna_change":"c.3982T>C","aliases":[],"variation_loc":[{"status":"current","assembly_name":"GRCh38","chr":"2","band":"2q24.3","start":"166009739","stop":"166009739","inner_start":"","inner_stop":"","outer_start":"","outer_stop":"","display_start":"166009739","display_stop":"166009739","assembly_acc_ver":"GCF_000001405.38","annotation_release":"","alt":"","ref":""},{"status":"previous","assembly_name":"GRCh37","chr":"2","band":"2q24.3","start":"166866249","stop":"166866249","inner_start":"","inner_stop":"","outer_start":"","outer_stop":"","display_start":"166866249","display_stop":"166866249","assembly_acc_ver":"GCF_000001405.25","annotation_release":"","alt":"","ref":""}],"allele_freq_set":[],"variant_type":"single nucleotide variant","canonical_spdi":"NC_000002.12:166009738:A:G","common_name":""}],"supporting_submissions":{"scv":["SCV005905783","SCV003524789"],"rcv":["RCV005254720","RCV006472763"]},"germline_classification":{"description":"Pathogenic/Likely pathogenic","last_evaluated":"2023/08/11 00:00","review_status":"criteria provided, multiple submitters, no conflicts","fda_recognized_database":"","trait_set":[{"trait_xrefs":[{"db_source":"Orphanet","db_id":"1934"},{"db_source":"MedGen","db_id":"C0393706"},{"db_source":"MONDO","db_id":"MONDO:0800491"}],"trait_name":"Early-infantile DEE"},{"trait_xrefs":[{"db_source":"Orphanet","db_id":"33069"},{"db_source":"MedGen","db_id":"C0751122"},{"db_source":"MONDO","db_id":"MONDO:0100135"},{"db_source":"OMIM","db_id":"607208"}],"trait_name":"Severe myoclonic epilepsy in infancy"}]},"clinical_impact_classification":{"description":"","last_evaluated":"1/01/01 00:00","review_status":"","fda_recognized_database":"","trait_set":[]},"oncogenicity_classification":{"description":"","last_evaluated":"1/01/01 00:00","review_status":"","fda_recognized_database":"","trait_set":[]},"record_status":"","gene_sort":"SCN1A","chr_sort":"02","location_sort":"000000000000166009739","variation_set_name":"","variation_set_id":"","genes":[{"symbol":"SCN1A","geneid":"6323","strand":"-","source":"submitted"},{"symbol":"LOC102724058","geneid":"102724058","strand":"+","source":"submitted"}],"molecular_consequence_list":["missense variant","non-coding transcript variant"],"protein_change":"S1317P, S1328P, S1316P, S514P, S1299P, S1300P","fda_recognized_database":""},"2169505":{"uid":"2169505","obj_type":"single nucleotide variant","accession":"VCV002169505","accession_version":"VCV002169505.5","title":"NM_001165963.4(SCN1A):c.3986G>C (p.Arg1329Pro)","variation_set":[{"measure_id":"1903452","variation_xrefs":[{"db_source":"ClinGen","db_id":"CA349052924"},{"db_source":"dbSNP","db_id":"1032633248"}],"variation_name":"NM_001165963.4(SCN1A):c.3986G>C 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00:00","review_status":"","fda_recognized_database":"","trait_set":[]},"record_status":"","gene_sort":"LOC102724058","chr_sort":"02","location_sort":"000000000000166009735","variation_set_name":"","variation_set_id":"","genes":[{"symbol":"LOC102724058","geneid":"102724058","strand":"+","source":"submitted"},{"symbol":"SCN1A","geneid":"6323","strand":"-","source":"submitted"}],"molecular_consequence_list":["missense variant","non-coding transcript variant"],"protein_change":"R1317P, R1300P, R1318P, R1301P, R515P, R1329P","fda_recognized_database":""},"1916226":{"uid":"1916226","obj_type":"single nucleotide variant","accession":"VCV001916226","accession_version":"VCV001916226.5","title":"NM_001165963.4(SCN1A):c.3986G>T (p.Arg1329Leu)","variation_set":[{"measure_id":"1977736","variation_xrefs":[{"db_source":"ClinGen","db_id":"CA349052922"},{"db_source":"dbSNP","db_id":"1032633248"}],"variation_name":"NM_001165963.4(SCN1A):c.3986G>T (p.Arg1329Leu)","cdna_change":"c.3986G>T","aliases":[],"variation_loc":[{"status":"current","assembly_name":"GRCh38","chr":"2","band":"2q24.3","start":"166009735","stop":"166009735","inner_start":"","inner_stop":"","outer_start":"","outer_stop":"","display_start":"166009735","display_stop":"166009735","assembly_acc_ver":"GCF_000001405.38","annotation_release":"","alt":"","ref":""},{"status":"previous","assembly_name":"GRCh37","chr":"2","band":"2q24.3","start":"166866245","stop":"166866245","inner_start":"","inner_stop":"","outer_start":"","outer_stop":"","display_start":"166866245","display_stop":"166866245","assembly_acc_ver":"GCF_000001405.25","annotation_release":"","alt":"","ref":""}],"allele_freq_set":[],"variant_type":"single nucleotide variant","canonical_spdi":"NC_000002.12:166009734:C:A","common_name":""}],"supporting_submissions":{"scv":["SCV002957851"],"rcv":["RCV006471542"]},"germline_classification":{"description":"Pathogenic","last_evaluated":"2022/09/06 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R1318L, R1329L, R515L, R1301L, R1300L","fda_recognized_database":""},"2430161":{"uid":"2430161","obj_type":"single nucleotide variant","accession":"VCV002430161","accession_version":"VCV002430161.2","title":"NM_001165963.4(SCN1A):c.3999G>A (p.Met1333Ile)","variation_set":[{"measure_id":"2403664","variation_xrefs":[{"db_source":"ClinGen","db_id":"CA349052845"},{"db_source":"dbSNP","db_id":"1559140155"}],"variation_name":"NM_001165963.4(SCN1A):c.3999G>A 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00:00","review_status":"criteria provided, single submitter","fda_recognized_database":"","trait_set":[{"trait_xrefs":[{"db_source":"MedGen","db_id":"C5543353"},{"db_source":"MONDO","db_id":"MONDO:0030268"},{"db_source":"OMIM","db_id":"619317"}],"trait_name":"Developmental and epileptic encephalopathy 6B"}]},"clinical_impact_classification":{"description":"","last_evaluated":"1/01/01 00:00","review_status":"","fda_recognized_database":"","trait_set":[]},"oncogenicity_classification":{"description":"","last_evaluated":"1/01/01 00:00","review_status":"","fda_recognized_database":"","trait_set":[]},"record_status":"","gene_sort":"LOC102724058","chr_sort":"02","location_sort":"000000000001660972","variation_set_name":"","variation_set_id":"","genes":[{"symbol":"LOC102724058","geneid":"102724058","strand":"+","source":"submitted"},{"symbol":"SCN1A","geneid":"6323","strand":"-","source":"submitted"}],"molecular_consequence_list":["missense variant","non-coding transcript variant"],"protein_change":"M1304I, M1305I, M1321I, M1322I, M1333I, M519I","fda_recognized_database":""}}}

图 D13　ClinVar（E-utilities esummary，无标签版）：高亮依次为 p.Ser1328Pro / Pathogenic/Likely pathogenic；p.Arg1329Pro / Pathogenic；p.Arg1329Leu / Pathogenic；p.Met1333Ile / Likely pathogenic

文件：screenshots/D13_ClinVar_邻近残基P_LP错义.jpg　|　来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2203183,2169505,1916226,2430161&retmode=json　|　访问：2026-11-05 00:20 UTC



图 D14　ClinVar（E-utilities esummary，无标签版）：SCN2A p.Glu1321Gln = Likely pathogenic；SCN8A p.Glu1311Val = Uncertain significance；两条的 review_status 都是 criteria provided, single submitter

文件：screenshots/D14_ClinVar_旁系同源对应位点.jpg ｜ 来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2036225,2842560&retmode=json ｜ 访问：2026-11-05 00:20 UTC

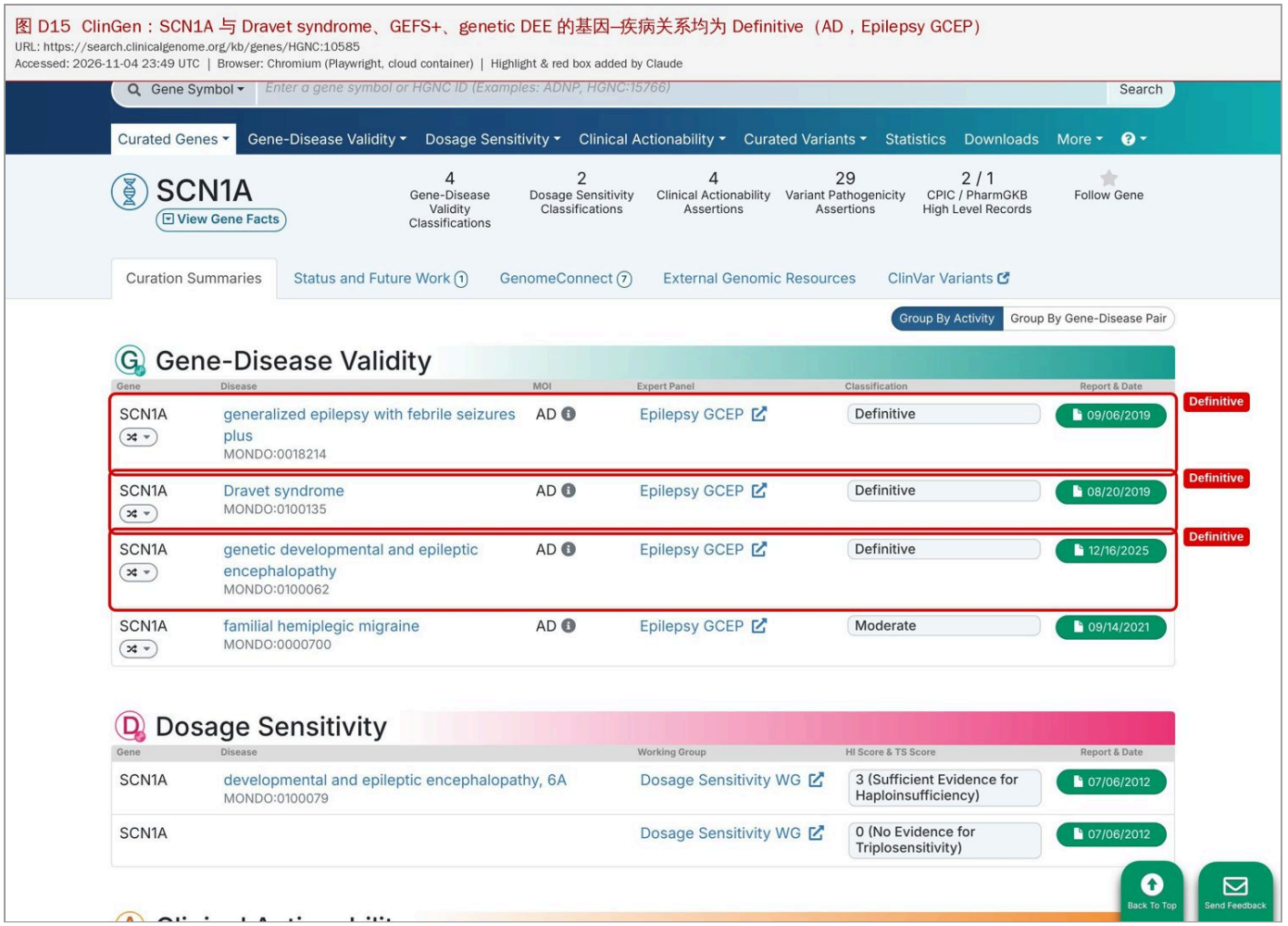


图 D15　ClinGen：SCN1A 与 Dravet syndrome、GEFS+、genetic DEE 的基因–疾病关系均为 Definitive（AD，Epilepsy GCEP）

文件：screenshots/D15_ClinGen_基因疾病有效性.jpg ｜ 来源：https://search.clinicalgenome.org/kb/genes/HGNC:10585 ｜ 访问：2026-11-04 23:49 UTC

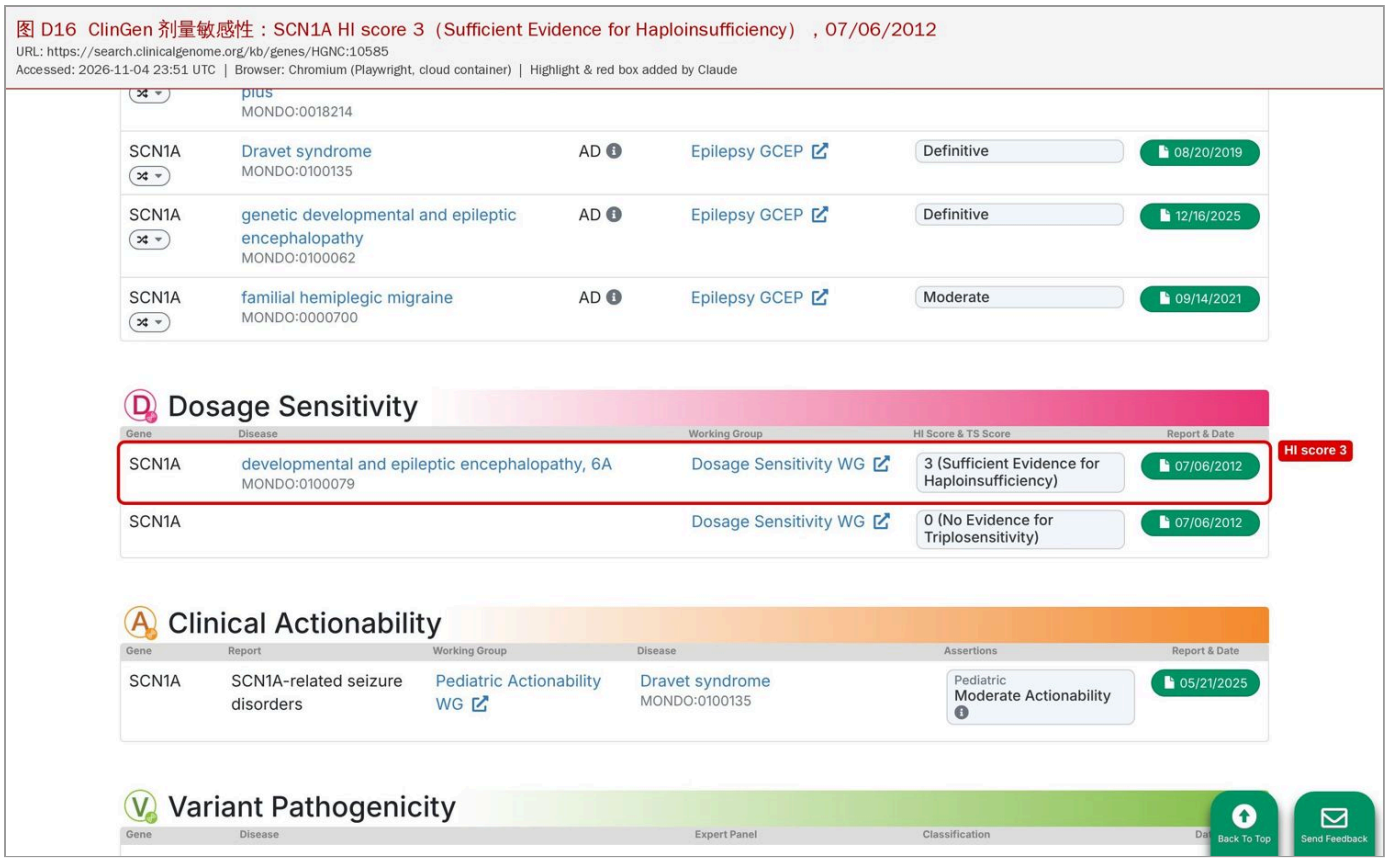


图 D16　ClinGen 剂量敏感性：SCN1A HI score 3（Sufficient Evidence for Haploinsufficiency），07/06/2012

文件：screenshots/D16_ClinGen_剂量敏感性HI3.jpg ｜ 来源：https://search.clinicalgenome.org/kb/genes/HGNC:10585 ｜ 访问：2026-11-04 23:51 UTC

D. 蛋白位置

图 D17 UniProt P35498 条目（纯文本版）：FT TRANSMEM 1314..1332 /note="Helical; Name=S4 of repeat III"；FT TOPO_DOM 1333..1345 为 Cytoplasmic。即 E1331 位于该跨膜螺旋末端，其后为胞质侧片段（我据此推断为 S4–S5 连接段）

URL: https://rest.uniprot.org/uniprotkb/P35498.txt

Accessed: 2026-11-04 23:52 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

```
FT                   /note="Cytoplasmic"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1219..1237
FT                   /note="Helical; Name=S1 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1238..1250
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1251..1276
FT                   /note="Helical; Name=S2 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1277..1278
FT                   /note="Cytoplasmic"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1279..1304
FT                   /note="Helical; Name=S3 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1305..1313
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1314..1332   residue 1331 is inside 1314–1332
FT                   /note="Helical; Name=S4 of repeat III"   DIII S4 (voltage sensor)
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1333..1345
FT                   /note="Cytoplasmic"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1346..1369
FT                   /note="Helical; Name=S5 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1370..1415
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   INTRAMEM        1416..1433
FT                   /note="Pore-forming"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1434..1457
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1458..1483
FT                   /note="Helical; Name=S6 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
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图 D17　UniProt P35498 条目（纯文本版）：FT TRANSMEM 1314..1332 /note="Helical; Name=S4 of repeat III"；FT TOPO_DOM 1333..1345 为 Cytoplasmic。即 E1331 位于该跨膜螺旋末端，其后为胞质侧片段（我据此推断为 S4–S5 连接段）

文件：screenshots/D17_UniProt_拓扑_DIII_S4.jpg ｜ 来源：https://rest.uniprot.org/uniprotkb/P35498.txt ｜ 访问：2026-11-04 23:52 UTC

E. 计算预测

p.(Glu1331Asp) 可由 c.3993A>T 或 c.3993A>C 产生，蛋白层面完全相同；两者的 REVEL、AlphaMissense 一致。Ensembl VEP 接口查询时返回 503/500，故改用 dbNSFP（D07）和 SpliceAI-lookup（D18–D21）。

REVEL 阈值（Pejaver 等 2022，表 2；VCEP 指定 REVEL）：PP3_Supporting ≥0.644，Moderate ≥0.773，Strong ≥0.932；BP4_Supporting ≤0.290，Moderate ≤0.183，Strong ≤0.016，Very Strong ≤0.003。0.613 落在 0.290–0.644 之间 → PP3 和 BP4 都不适用。阈值是通过 PMC 开放获取全文接口（PMC9748256）读出来核对的；PMC 网页对容器浏览器弹验证码，【需查】原文表 2 截图请在你的 Chrome 补。SpliceAI-lookup 页面的 Points 列对 REVEL 0.613 记 0 分（D19、D21）；该网站的计分依据我没有核对，所以只作参考。

注意：旧版 D21 是在页面 REVEL 尚未加载时截的，当时页面写着 REVEL “not available”，我的标签却写了 0.613；旧版 D19 的标签盖住了分数本身。两张都已作废，现为等 REVEL 加载完成后重拍、不加标签的版本。

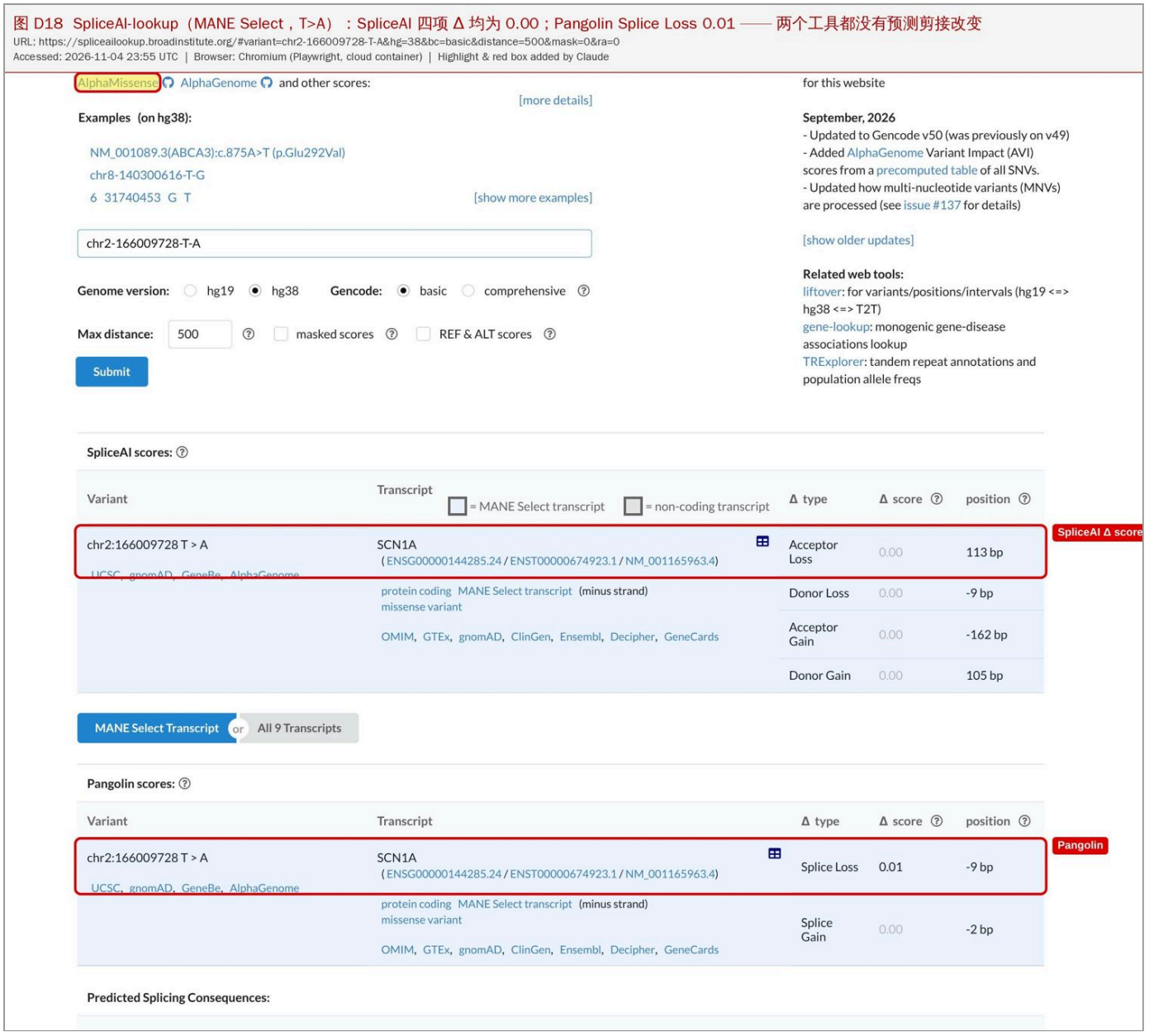


图 D18　SpliceAI-lookup（MANE Select，T>A）：SpliceAI 四项 Δ 均为 0.00；Pangolin Splice Loss 0.01 —— 两个工具都没有预测剪接改变

文件：screenshots/D18_SpliceAI_TA_剪接预测.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-A&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:55 UTC

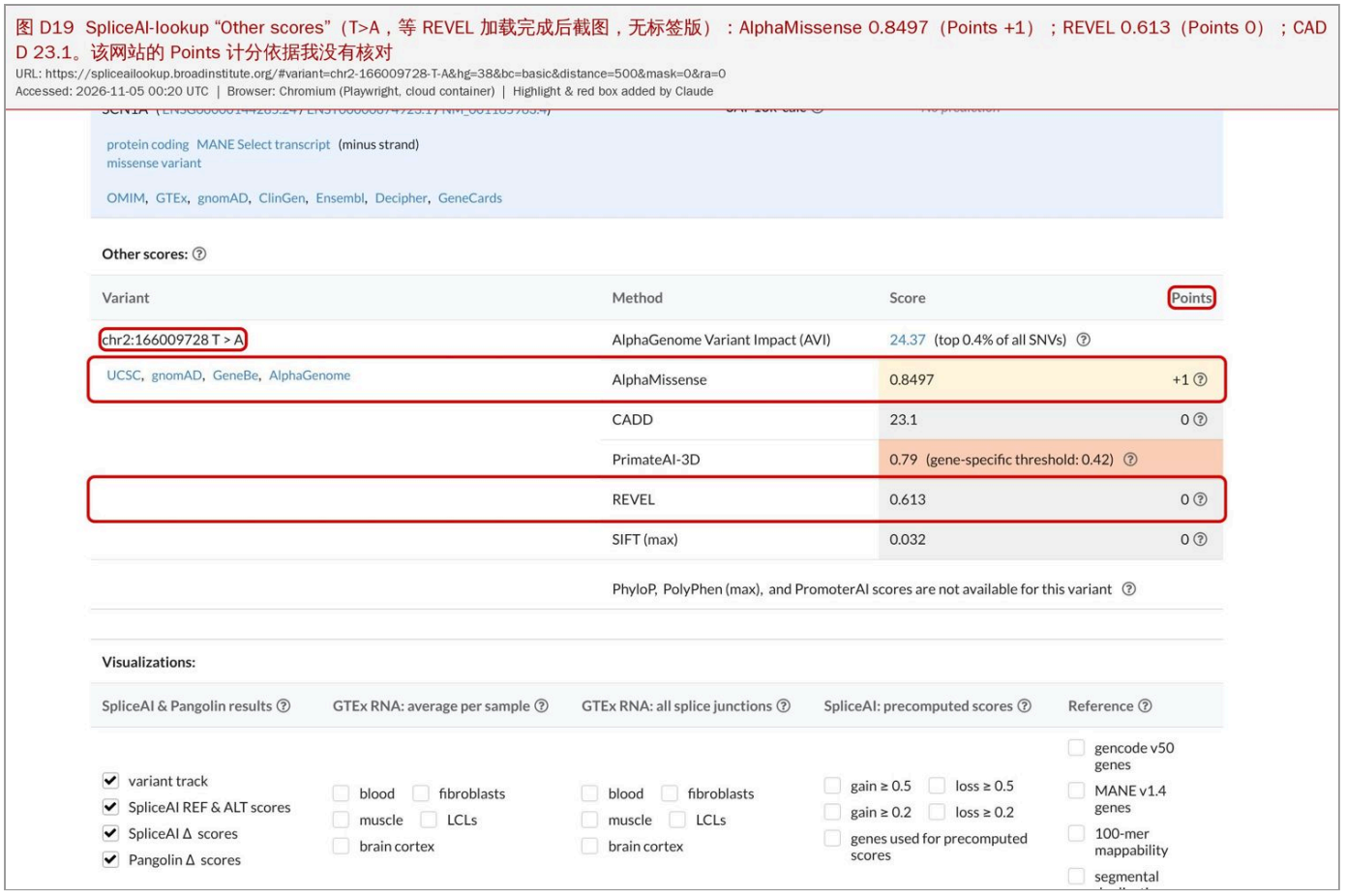


图 D19　SpliceAI-lookup “Other scores”（T>A，等 REVEL 加载完成后截图，无标签版）：AlphaMissense 0.8497（Points +1）；REVEL 0.613（Points 0）；CADD 23.1。该网站的 Points 计分依据我没有核对

文件：screenshots/D19_SpliceAI_TA_REVEL_AlphaMissense.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-A&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-05 00:20 UTC

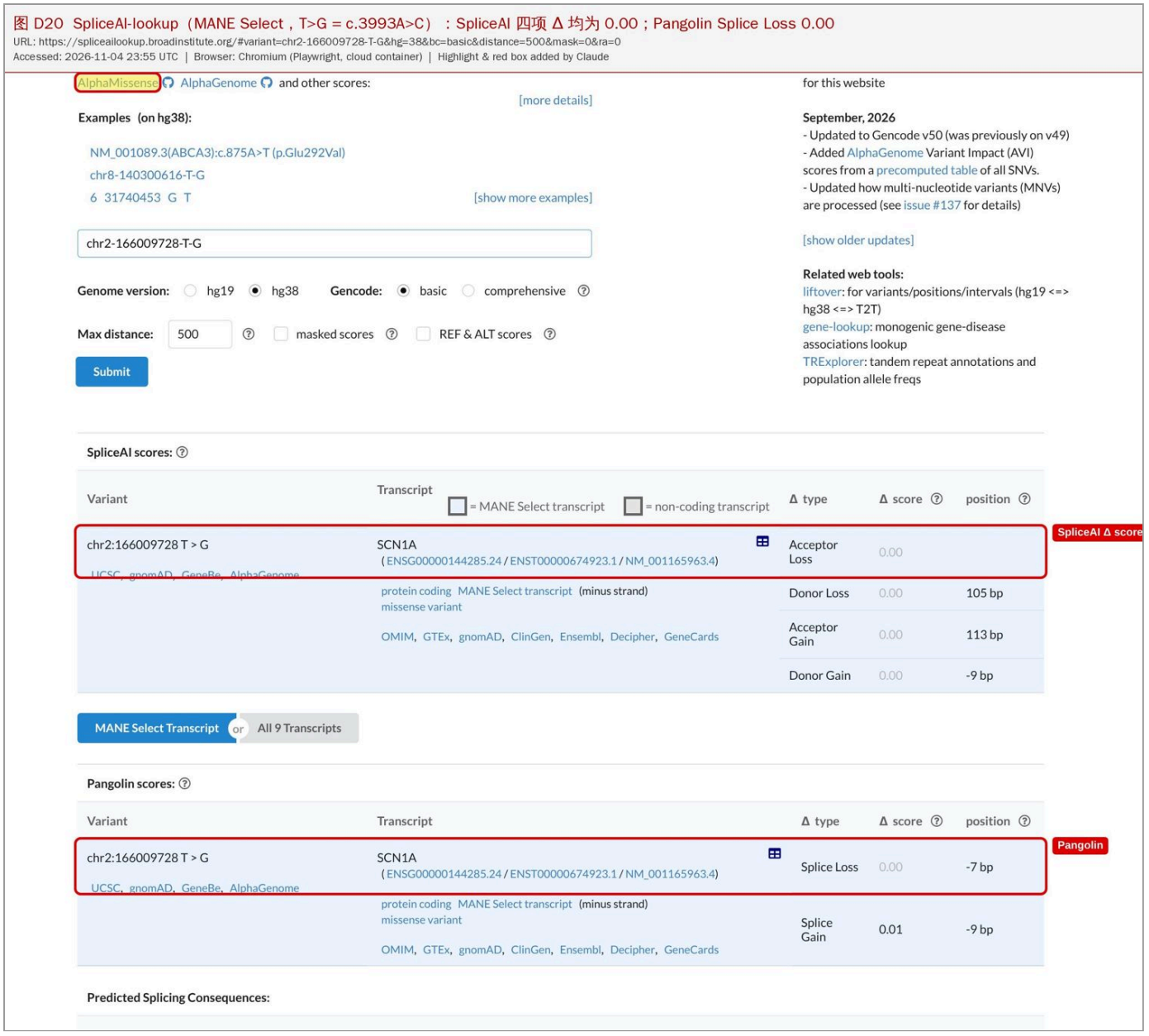


图 D20　SpliceAI-lookup（MANE Select，T>G = c.3993A>C）：SpliceAI 四项 Δ 均为 0.00；Pangolin Splice Loss 0.00

文件：screenshots/D20_SpliceAI_TG_剪接预测.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-G&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:55 UTC

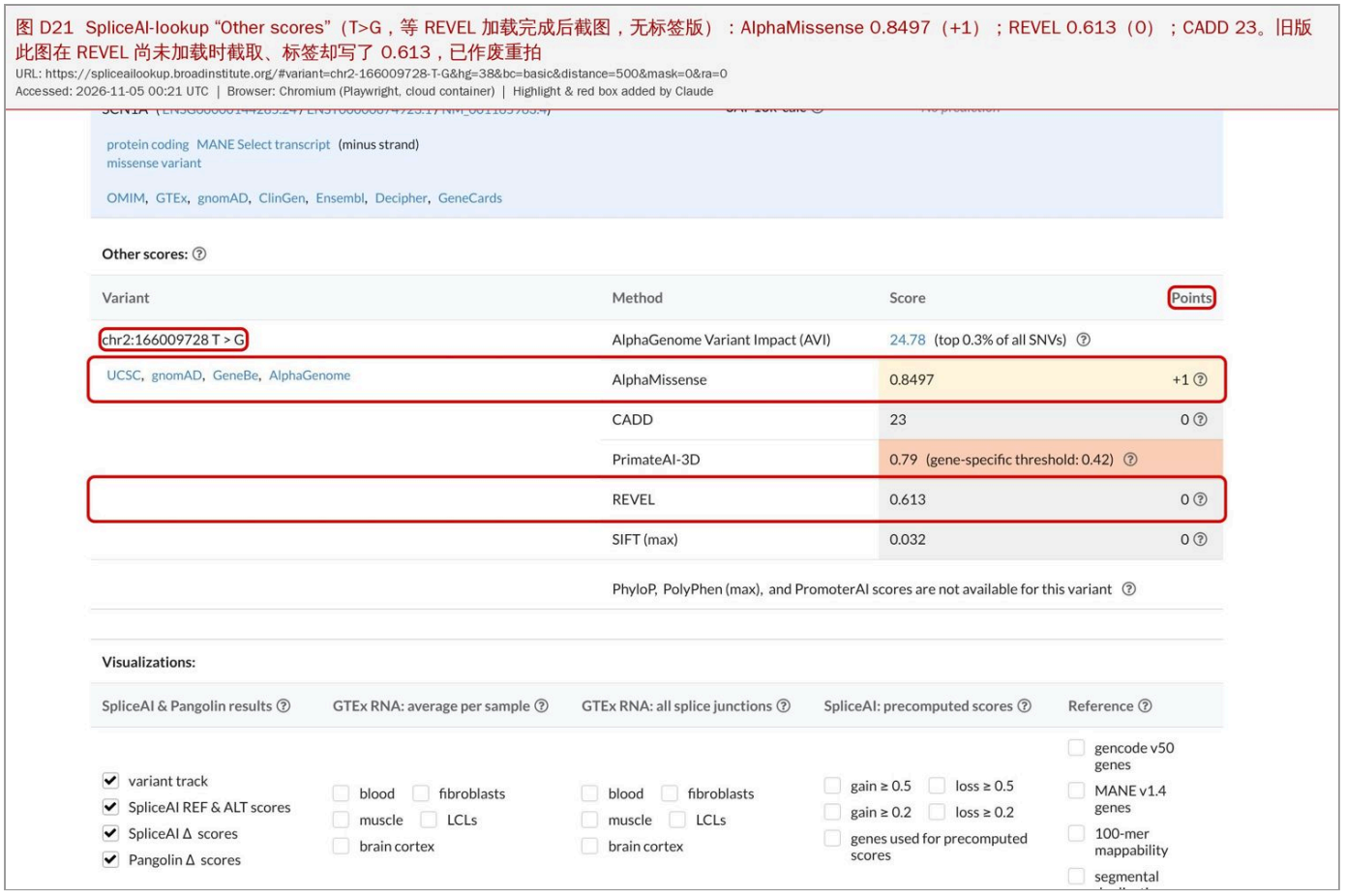


图 D21　SpliceAI-lookup “Other scores”（T>G，等 REVEL 加载完成后截图，无标签版）：AlphaMissense 0.8497（+1）；REVEL 0.613（0）；CADD 23。旧版此图在 REVEL 尚未加载时截取、标签却写了 0.613，已作废重拍

文件：screenshots/D21_SpliceAI_TG_REVEL_AlphaMissense.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-G&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-05 00:21 UTC

保守性：手动比对（表 A1）

方法：从 UniProt REST 取序列，以人 SCN1A 1311–1351 为查询序列，用 Smith–Waterman 局部比对（BLOSUM62，线性空位罚分 −4，我自己写的脚本）定位对应残基。这是手动比对，不是数据库现成结果；UCSC 比对/phyloP 页面查询时超时，未能截图（dbNSFP 给出的 phyloP100way = 0.615，对应摆动位的核苷酸保守性，意义有限）。黄色列 = 人 SCN1A 第 1331 位。

物种 / 基因	UniProt	比对窗口（人 1311–1351）	对应残基	位置
Human SCN1A	P35498 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Mouse Scn1a	A2APX8 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Rat Scn1a	P04774 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Cow SCN1A	M0QW12 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Chicken SCN1A	A0A8V1A8W9 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1338
Xenopus tropicalis scn1a	A0A8J1IT78 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALIGAIPSIMNVL	E	1329
Zebrafish scn1lab	F1QXF5 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1312
Zebrafish scn1laa	Q20JQ9 (TrEMBL)	AIKSLRTLRAFRPLRALSRFD**D**GMRVVVNALLGAIPSIMNVL	D	1280
Drosophila para	P35500 (Swiss-Prot)	AFKTMRTLRALRPLRAMSRM**Q**GMRVVVNALVQAIPSIFNVL	Q	1414
Human SCN2A	Q99250 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1321

Human SCN3A	Q9NY46 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFEGMRVVVNALVGAIPSIMNVL	E	1319
Human SCN8A	Q9UQD0 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFEGMRVVVNALVGAIPSIMNVL	E	1311
Human SCN4A	P35499 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALLGAIPSIMNVL	E	1144
Human SCN5A	Q14524 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALVGAIPSIMNVL	E	1318
Human SCN9A	Q15858 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALIGAIPSIMNVL	E	1305
Human SCN10A	Q9Y5Y9 (Swiss-Prot)	-IKALRTLRALRPLRALSRFEGMRVVVDALVGAIPSIMNVL	E	1265
Human SCN11A	Q9UI33 (Swiss-Prot)	-LKSFRTLRALRPLRALSQFEGMKVVVNALIGAIPAILNVL	E	1162

补充：UniProt 中斑马鱼 scn1laa 共 5 个条目（Q20JQ9、A0A8M1PAT7、A0ACD6B7I9、A0AB32TV97、A0AC58GDP1），在该位置都是 D。它们都是未经人工审核的 TrEMBL 条目，而且来自同一个基因，不能算 5 条独立证据。另外核对了 ClinVar 标题所用的旁系同源转录本（SCN2A NM_001040142.2、SCN3A NM_006922.4、SCN8A NM_001330260.2），对应位点都是 E。

解读（推断）：Glu 在人类全部 9 个电压门控钠通道 α 亚基和所查四足动物中保守，提示该位点可能有功能重要性；斑马鱼 scn1laa 为 Asp，提示 E→D 在鱼类同源通道中可能可以耐受，但鱼类通道的序列背景不同，不能直接外推到人 Nav1.1。写进报告时用 suggests / may，并注明是手动比对。

F. ClinGen Epilepsy Sodium Channel VCEP《SCN1A 细则》v2.0（GN067）

附件 PM1 Table、Paralogous Gene Table、PVS1 exon numbering 为 xlsx，在页面 Files & Images 点击下载按钮获得（保存在 ClinGen_VCEP附件/），D28、D29 是本地渲染后的截图，数值未改动。附件 “Combining Rules”（VCEP 的积分合并规则）和 “PS1_Variants impacting splicing” 是 PDF，按“PDF 只在你要求时下载”的规则没有下载。



图 D22　ClinGen CSpec GN067：Epilepsy Sodium Channel VCEP《SCN1A 细则》Version 2.0，Released 1/7/2025，关键词中的转录本为 NM_001165963.3。NM_001165963.3 与 .4 都编码 NP_001159435.1、序列相同（E-utilities GenBank 记录脚本核对，未截图）

文件：screenshots/D22_ClinGen_VCEP_SCN1A细则v2.0.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

图 D23 VCEP 细则（无标签版）：PM1 Moderate = 位于 Pathogenic Enriched Region（PER 定义：跨基因家族富集致病变异、缺少 gnomAD 变异）；PM2 Supporting = ≥10,000 个等位基因中 ≤1 个
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:20 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

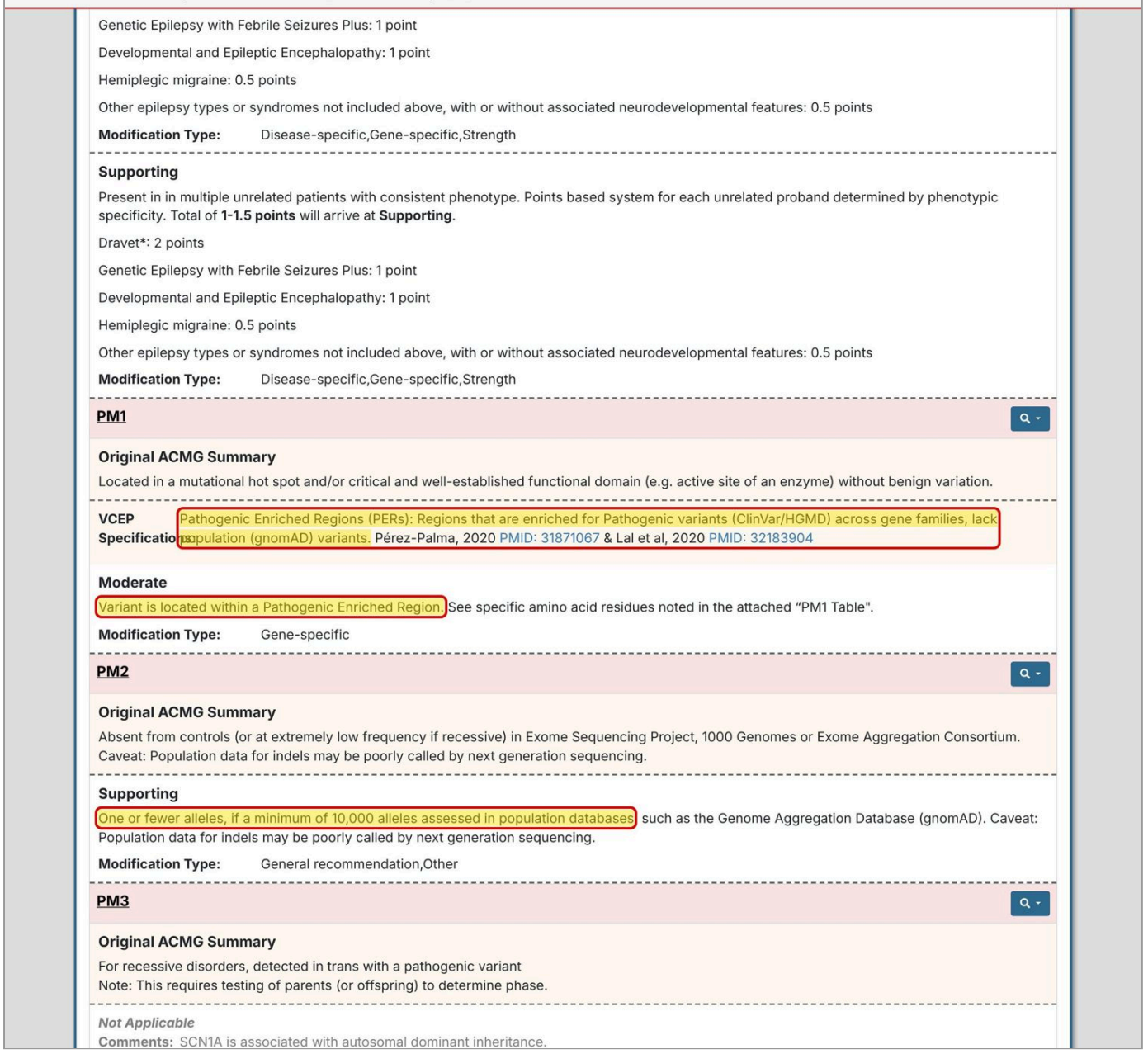


图 D23　VCEP 细则（无标签版）：PM1 Moderate = 位于 Pathogenic Enriched Region（PER 定义：跨基因家族富集致病变异、缺少 gnomAD 变异）；PM2 Supporting = ≥10,000 个等位基因中 ≤1 个

文件：screenshots/D23_VCEP_PM1_PM2.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-05 00:20 UTC

图 D24 VCEP 细则：PP2 不适用（良性错义常见）；PP3 用 REVEL 按 Pejaver 2022（PMID 36413997）判定，最高 Moderate，PP3+PM1 合计不超过 Strong；PP4 不适用（表型特异性已并入 PS2/PM6/PS4）
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-04 23:53 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

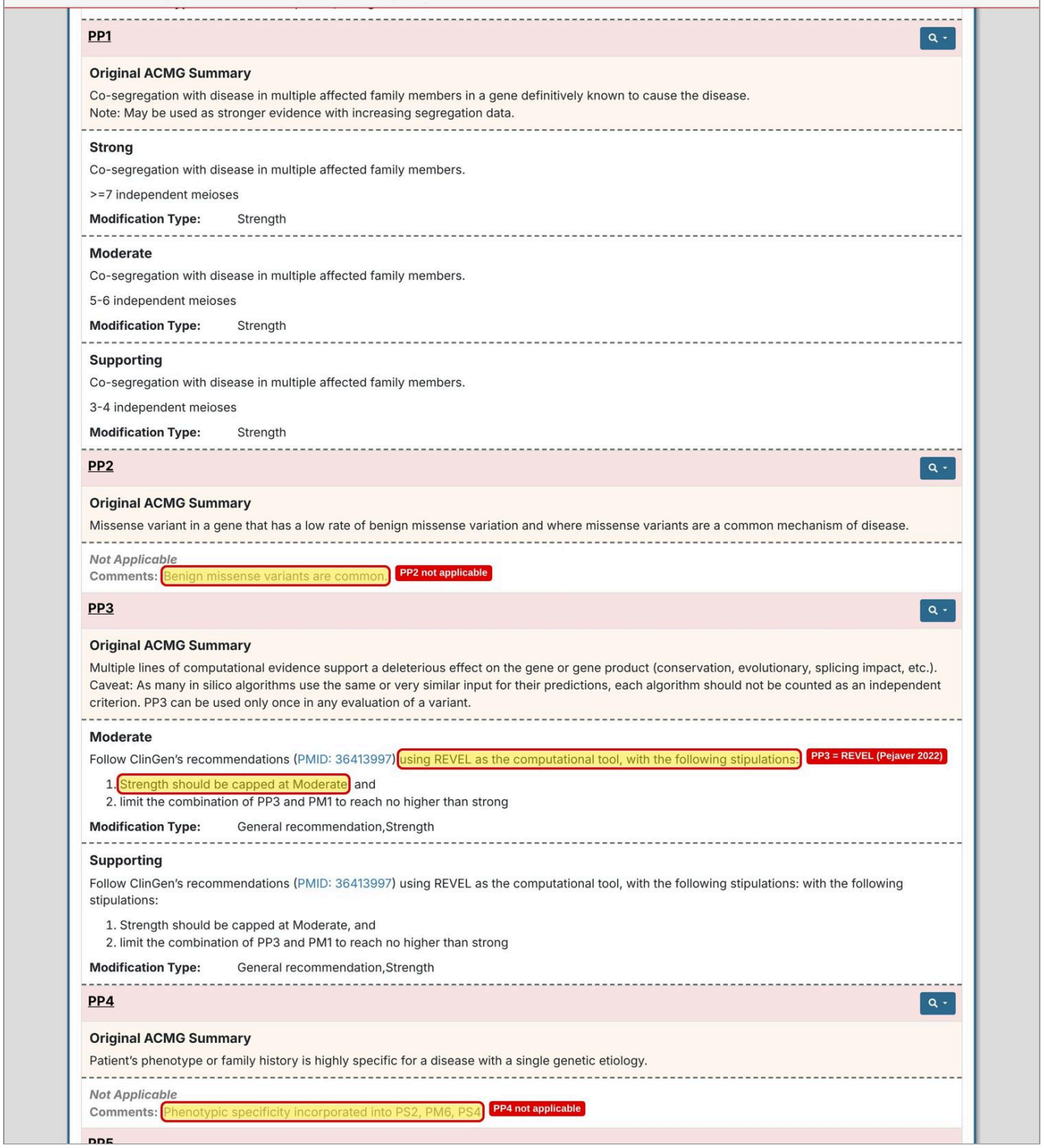


图 D24　VCEP 细则：PP2 不适用（良性错义常见）；PP3 用 REVEL 按 Pejaver 2022（PMID 36413997）判定，最高 Moderate，PP3+PM1 合计不超过 Strong；PP4 不适用（表型特异性已并入 PS2/PM6/PS4）

文件：screenshots/D24_VCEP_PP2_PP3_PP4.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

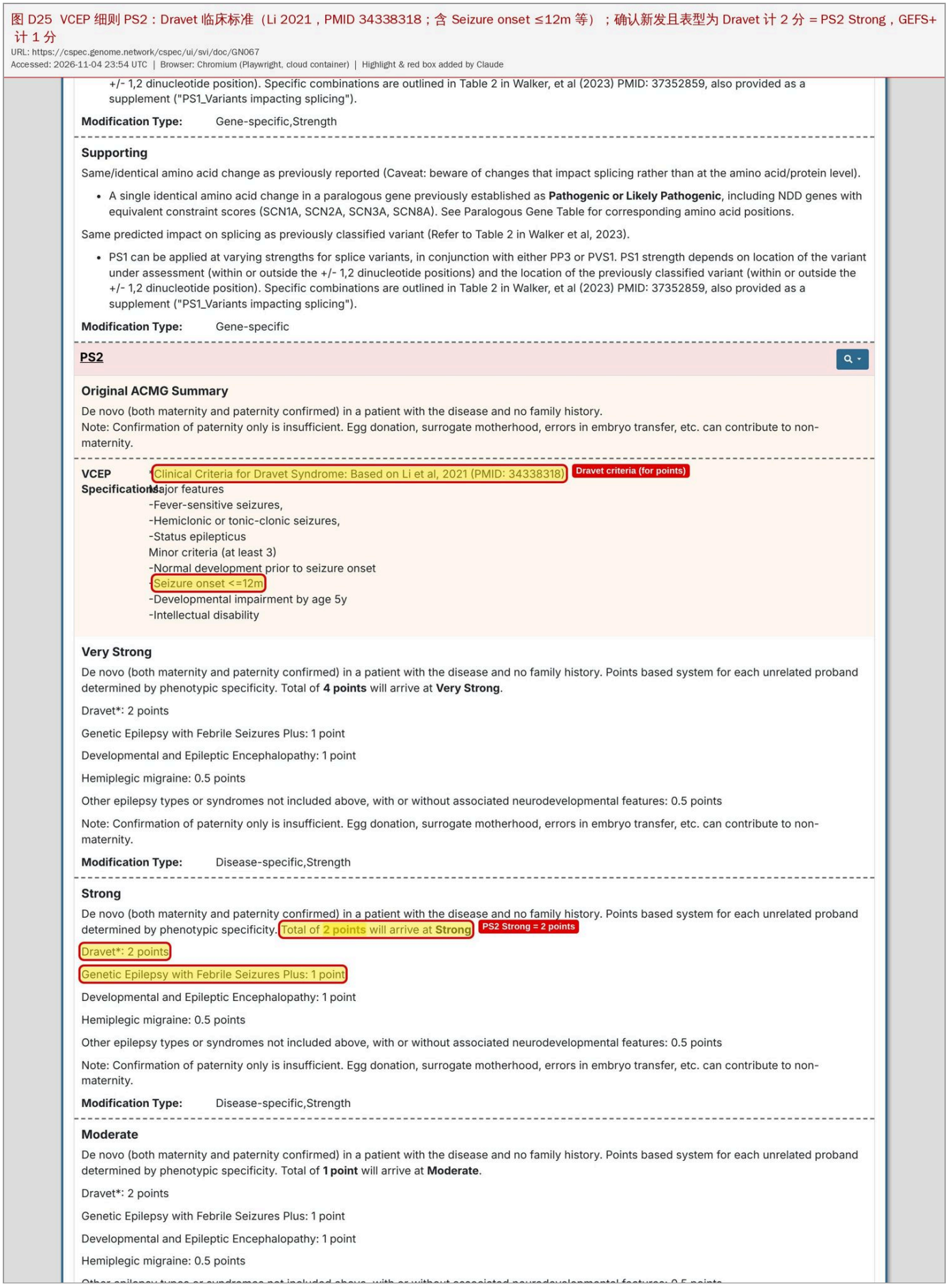


图 D25　VCEP 细则 PS2：Dravet 临床标准（Li 2021，PMID 34338318；含 Seizure onset ≤12m 等）；确认新发且表型为 Dravet 计 2 分 = PS2 Strong，GEFS+ 计 1 分

文件：screenshots/D25_VCEP_PS2_Dravet标准与积分.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:54 UTC

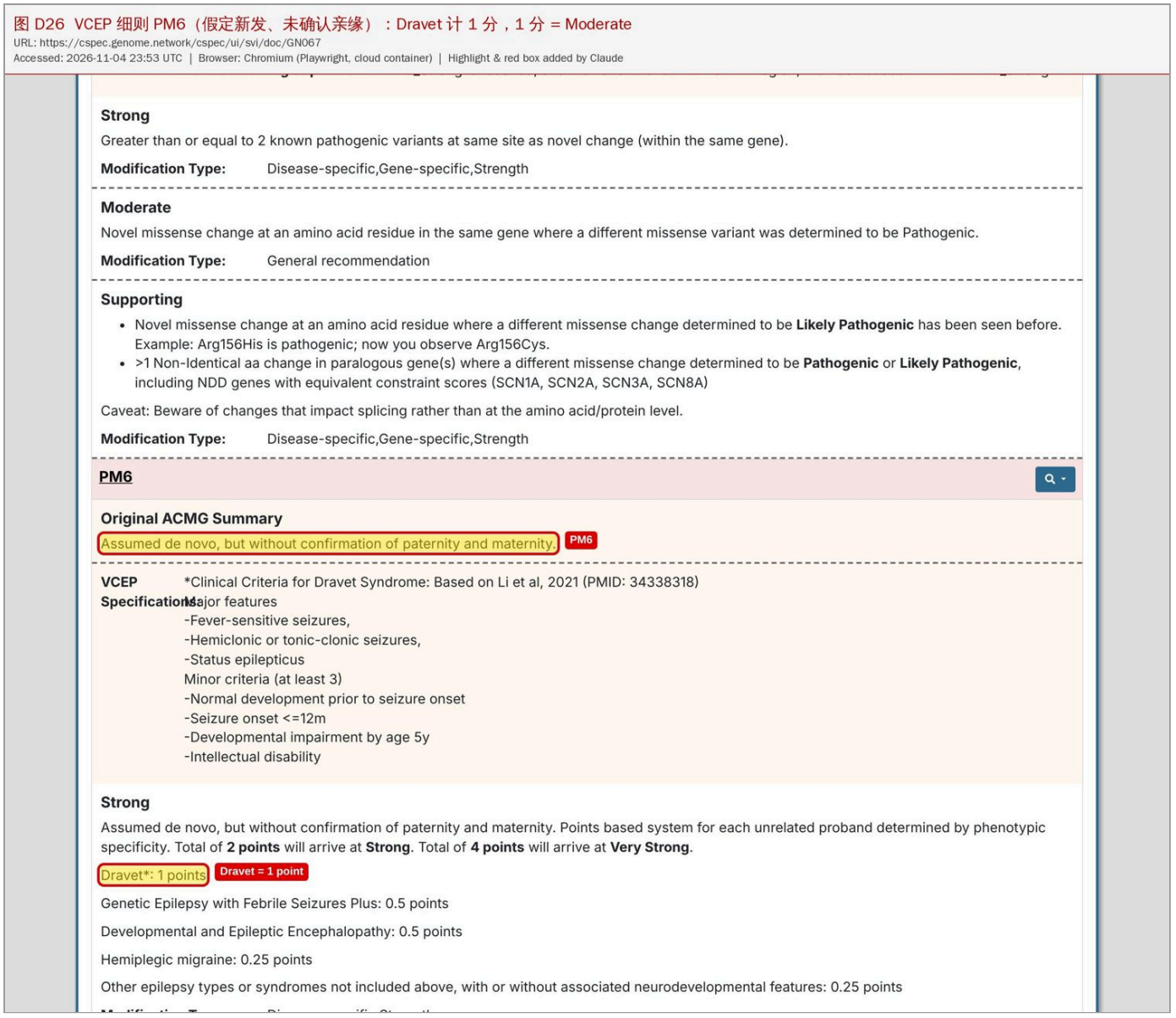
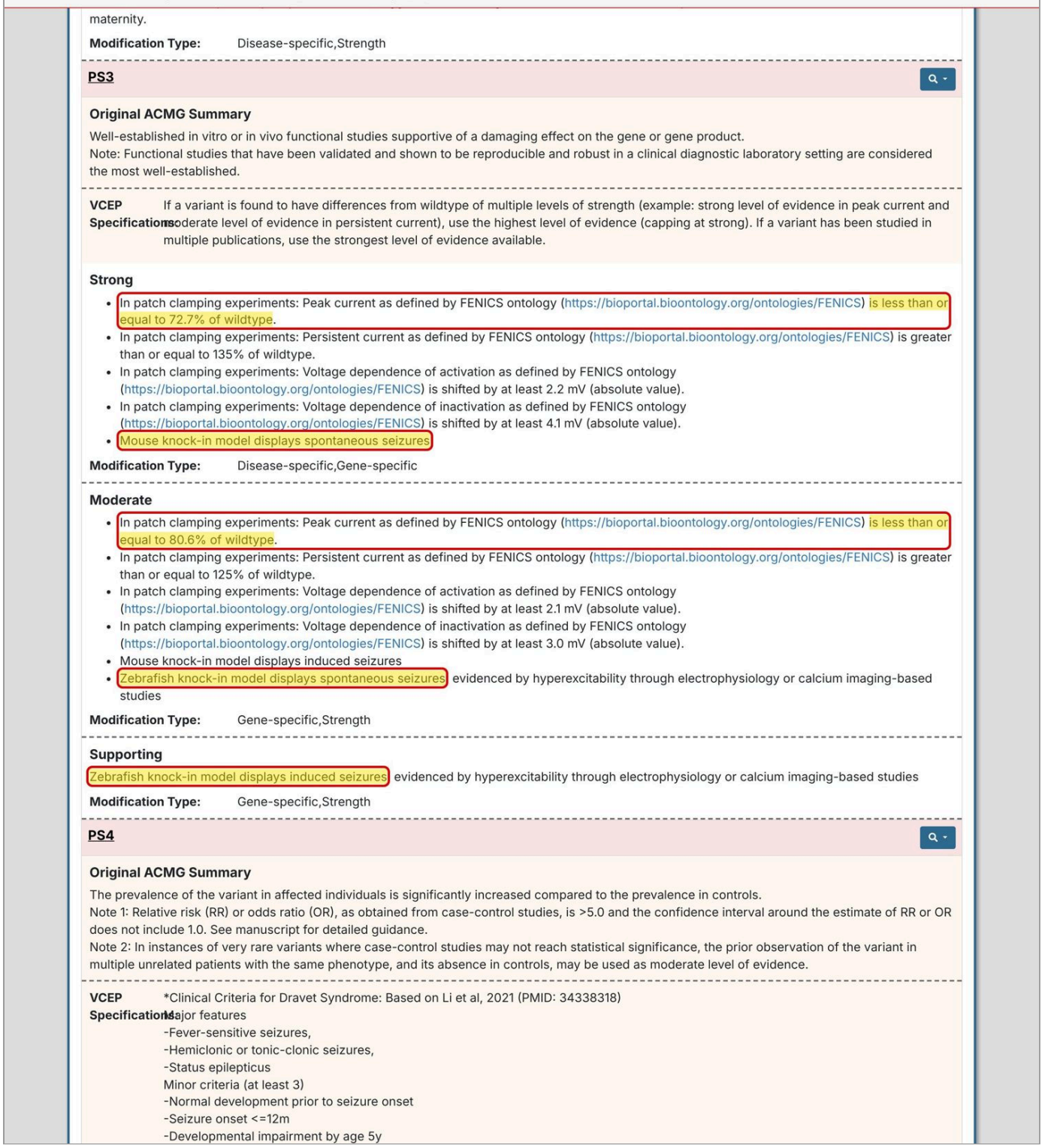


图 D26　VCEP 细则 PM6（假定新发、未确认亲缘）：Dravet 计 1 分，1 分 = Moderate

文件：screenshots/D26_VCEP_PM6积分.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

图 D27 VCEP 细则 PS3（无标签版）：Strong = 膜片钳峰电流 ≤72.7% 野生型，或小鼠敲入自发惊厥；Moderate = 峰电流 ≤80.6%，或斑马鱼敲入自发惊厥；Supporting = 斑马鱼敲入诱发惊厥
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:23 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude


maternity.

Modification Type: Disease-specific,Strength

PS3

Original ACMG Summary

Well-established in vitro or in vivo functional studies supportive of a damaging effect on the gene or gene product.
Note: Functional studies that have been validated and shown to be reproducible and robust in a clinical diagnostic laboratory setting are considered the most well-established.

VCEP Specifications: If a variant is found to have differences from wildtype of multiple levels of strength (example: strong level of evidence in peak current and moderate level of evidence in persistent current), use the highest level of evidence (capping at strong). If a variant has been studied in multiple publications, use the strongest level of evidence available.

Strong

- In patch clamping experiments: Peak current as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is less than or equal to 72.7% of wildtype.
- In patch clamping experiments: Persistent current as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is greater than or equal to 135% of wildtype.
- In patch clamping experiments: Voltage dependence of activation as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is shifted by at least 2.2 mV (absolute value).
- In patch clamping experiments: Voltage dependence of inactivation as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is shifted by at least 4.1 mV (absolute value).
- Mouse knock-in model displays spontaneous seizures

Modification Type: Disease-specific,Gene-specific

Moderate

- In patch clamping experiments: Peak current as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is less than or equal to 80.6% of wildtype.
- In patch clamping experiments: Persistent current as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is greater than or equal to 125% of wildtype.
- In patch clamping experiments: Voltage dependence of activation as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is shifted by at least 2.1 mV (absolute value).
- In patch clamping experiments: Voltage dependence of inactivation as defined by FENICS ontology (https://bioportal.bioontology.org/ontologies/FENICS) is shifted by at least 3.0 mV (absolute value).
- Mouse knock-in model displays induced seizures
- Zebrafish knock-in model displays spontaneous seizures evidenced by hyperexcitability through electrophysiology or calcium imaging-based studies

Modification Type: Gene-specific,Strength

Supporting

Zebrafish knock-in model displays induced seizures evidenced by hyperexcitability through electrophysiology or calcium imaging-based studies

Modification Type: Gene-specific,Strength

PS4

Original ACMG Summary

The prevalence of the variant in affected individuals is significantly increased compared to the prevalence in controls.
Note 1: Relative risk (RR) or odds ratio (OR), as obtained from case-control studies, is >5.0 and the confidence interval around the estimate of RR or OR does not include 1.0. See manuscript for detailed guidance.
Note 2: In instances of very rare variants where case-control studies may not reach statistical significance, the prior observation of the variant in multiple unrelated patients with the same phenotype, and its absence in controls, may be used as moderate level of evidence.

VCEP Specifications: *Clinical Criteria for Dravet Syndrome: Based on Li et al, 2021 (PMID: 34338318)
Major features
-Fever-sensitive seizures,
-Hemiclonic or tonic-clonic seizures,
-Status epilepticus
Minor criteria (at least 3)
-Normal development prior to seizure onset
-Seizure onset <=12m
-Developmental impairment by age 5y

图 D27　VCEP 细则 PS3（无标签版）：Strong = 膜片钳峰电流 ≤72.7% 野生型，或小鼠敲入自发惊厥；Moderate = 峰电流 ≤80.6%，或斑马鱼敲入自发惊厥；Supporting = 斑马鱼敲入诱发惊厥

文件：screenshots/D27_VCEP_PS3功能实验阈值.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-05 00:23 UTC

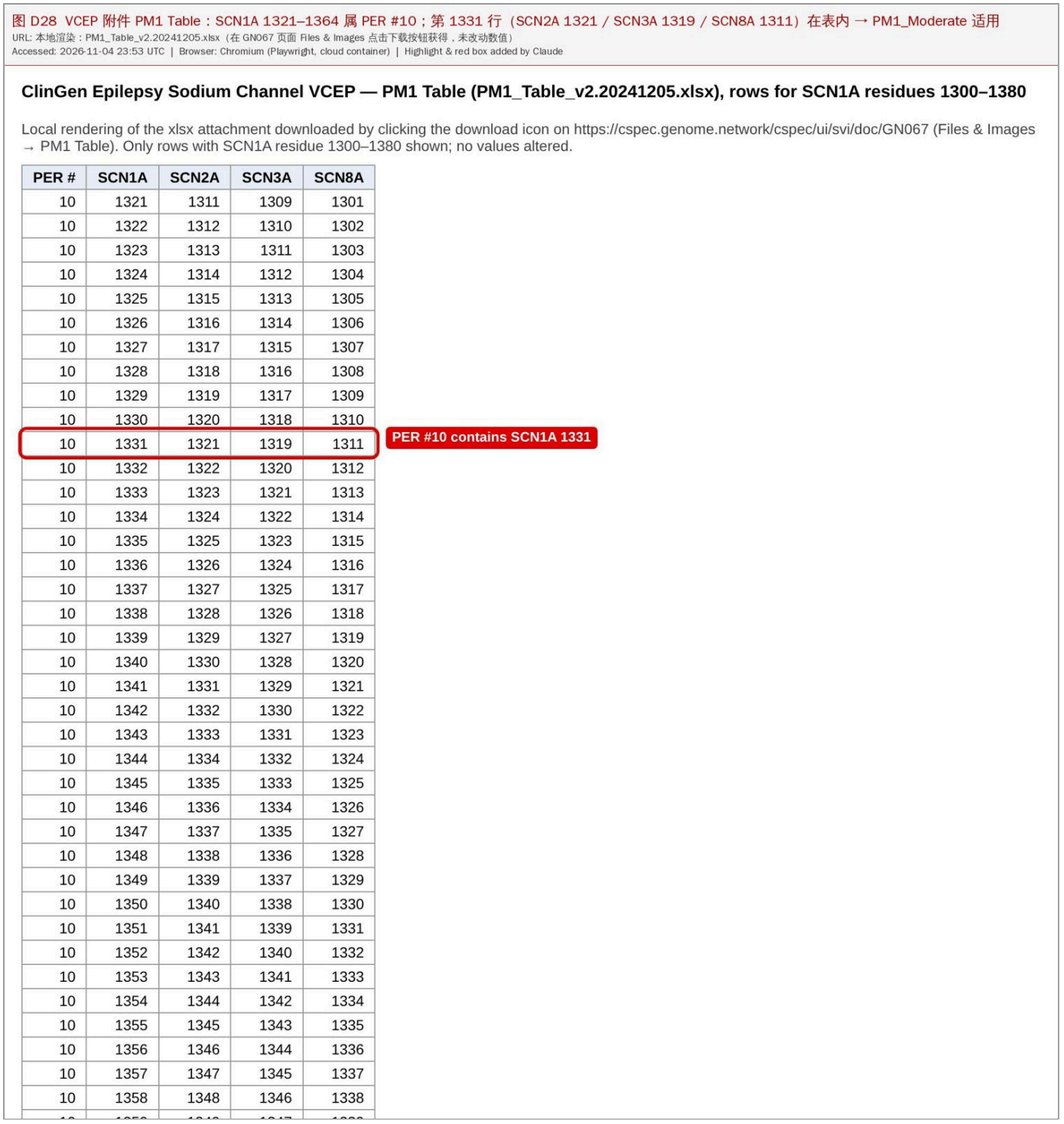


图 D28 VCEP 附件 PM1 Table：SCN1A 1321–1364 属 PER #10；第 1331 行（SCN2A 1321 / SCN3A 1319 / SCN8A 1311）在表内 → PM1_Moderate 适用
URL: 本地渲染：PM1_Table_v2.20241205.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值）
Accessed: 2026-11-04 23:53 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

ClinGen Epilepsy Sodium Channel VCEP — PM1 Table (PM1_Table_v2.20241205.xlsx), rows for SCN1A residues 1300–1380

Local rendering of the xlsx attachment downloaded by clicking the download icon on https://cspec.genome.network/cspec/ui/svi/doc/GN067 (Files & Images → PM1 Table). Only rows with SCN1A residue 1300–1380 shown; no values altered.

PER #	SCN1A	SCN2A	SCN3A	SCN8A
10	1321	1311	1309	1301
10	1322	1312	1310	1302
10	1323	1313	1311	1303
10	1324	1314	1312	1304
10	1325	1315	1313	1305
10	1326	1316	1314	1306
10	1327	1317	1315	1307
10	1328	1318	1316	1308
10	1329	1319	1317	1309
10	1330	1320	1318	1310
10	1331	1321	1319	1311
10	1332	1322	1320	1312
10	1333	1323	1321	1313
10	1334	1324	1322	1314
10	1335	1325	1323	1315
10	1336	1326	1324	1316
10	1337	1327	1325	1317
10	1338	1328	1326	1318
10	1339	1329	1327	1319
10	1340	1330	1328	1320
10	1341	1331	1329	1321
10	1342	1332	1330	1322
10	1343	1333	1331	1323
10	1344	1334	1332	1324
10	1345	1335	1333	1325
10	1346	1336	1334	1326
10	1347	1337	1335	1327
10	1348	1338	1336	1328
10	1349	1339	1337	1329
10	1350	1340	1338	1330
10	1351	1341	1339	1331
10	1352	1342	1340	1332
10	1353	1343	1341	1333
10	1354	1344	1342	1334
10	1355	1345	1343	1335
10	1356	1346	1344	1336
10	1357	1347	1345	1337
10	1358	1348	1346	1338

图 D28　VCEP 附件 PM1 Table：SCN1A 1321–1364 属 PER #10；第 1331 行（SCN2A 1321 / SCN3A 1319 / SCN8A 1311）在表内 → PM1_Moderate 适用

文件：screenshots/D28_VCEP附件_PM1表_PER10.jpg ｜ 来源：本地渲染：PM1_Table_v2.20241205.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值） ｜ 访问：2026-11-04 23:53 UTC

图 D29 VCEP 附件 旁系同源氨基酸对照表：SCN1A 1331 = SCN2A 1321 = SCN3A 1319 = SCN8A 1311（用于 PS1/PM5 的旁系同源条款）
URL: 本地渲染：Paralogous-Genes_AminoAcid-Mapping.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值）
Accessed: 2026-11-04 23:53 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

ClinGen Epilepsy Sodium Channel VCEP — Paralogous Gene Table: Amino Acid Mapping (Paralogous-Genes_AminoAcid-Mapping.xlsx), SCN1A residues 1321–1341

Local rendering of the xlsx attachment downloaded by clicking the download icon on https://cspec.genome.network/cspec/ui/svi/doc/GN067 (Files & Images → Paralogous Gene Table). Only rows with SCN1A residue 1321–1341 shown; no values altered.

SCN1A	SCN2A	SCN3A	SCN8A
1321	1311	1309	1301
1322	1312	1310	1302
1323	1313	1311	1303
1324	1314	1312	1304
1325	1315	1313	1305
1326	1316	1314	1306
1327	1317	1315	1307
1328	1318	1316	1308
1329	1319	1317	1309
1330	1320	1318	1310
1331	1321	1319	1311
1332	1322	1320	1312
1333	1323	1321	1313
1334	1324	1322	1314
1335	1325	1323	1315
1336	1326	1324	1316
1337	1327	1325	1317
1338	1328	1326	1318
1339	1329	1327	1319
1340	1330	1328	1320
1341	1331	1329	1321

SCN1A 1331 = SCN2A 1321 = SCN3A 1319 = SCN8A 1311

图 D29　VCEP 附件 旁系同源氨基酸对照表：SCN1A 1331 = SCN2A 1321 = SCN3A 1319 = SCN8A 1311（用于 PS1/PM5 的旁系同源条款）

文件：screenshots/D29_VCEP附件_旁系同源对照表.jpg ｜ 来源：本地渲染：Paralogous-Genes_AminoAcid-Mapping.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值） ｜ 访问：2026-11-04 23:53 UTC

F2　2026-11-05 补截：此前只读过接口文本、没有截图的条款

图 D30　VCEP 细则 PS1：Strong = 同一氨基酸改变曾判 Pathogenic，或旁系同源基因中 >1 个相同改变判 P/LP；Supporting = 旁系同源中 1 个相同改变判 P/LP
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:21 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

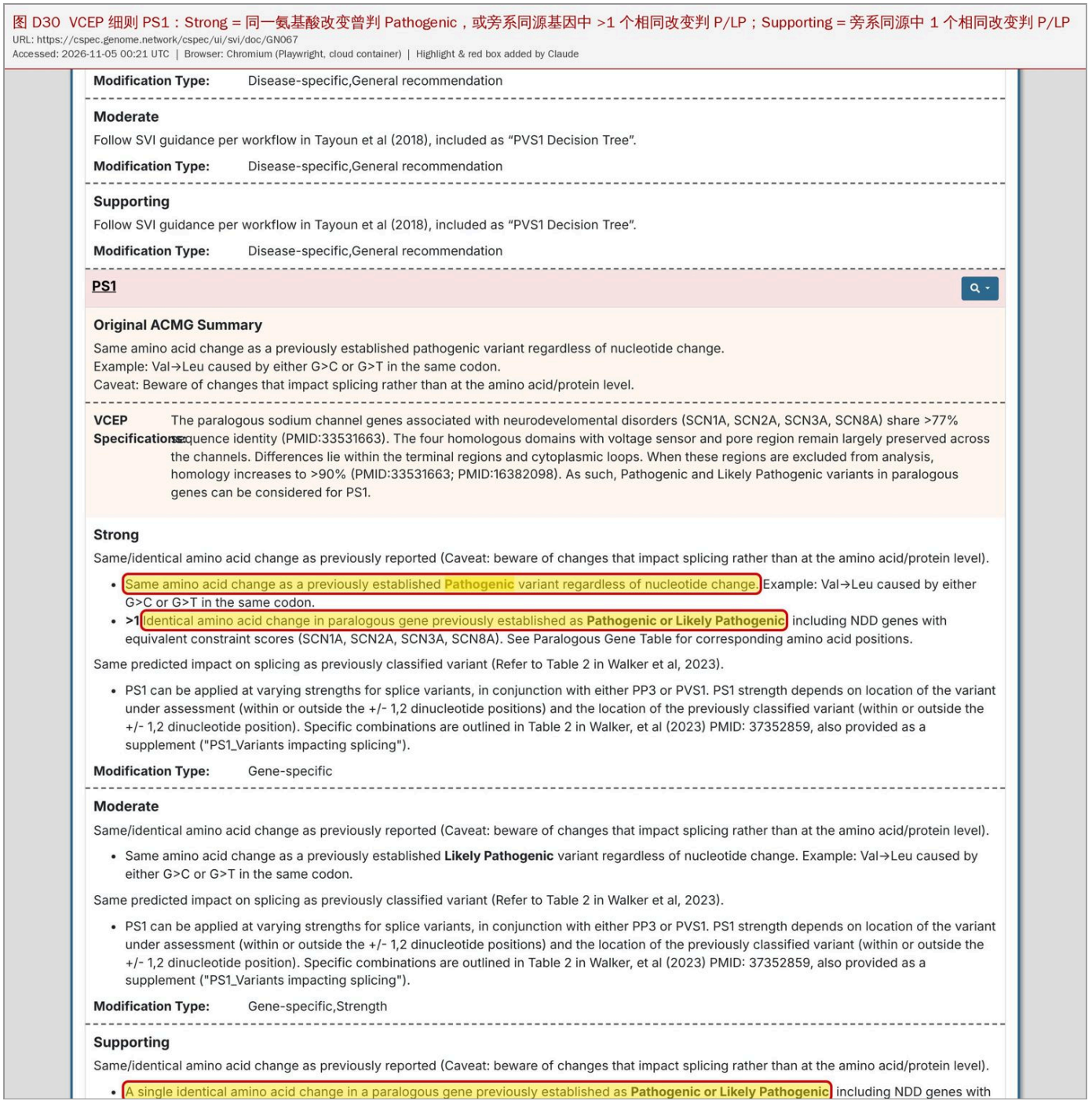


图 D30　VCEP 细则 PS1：Strong = 同一氨基酸改变曾判 Pathogenic，或旁系同源基因中 >1 个相同改变判 P/LP；Supporting = 旁系同源中 1 个相同改变判 P/LP

文件：screenshots/D30_VCEP_PS1.jpg　|　来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067　|　访问：2026-11-05 00:21 UTC

图 D31 VCEP 细则 PM5：Strong = 同位点 ≥2 个已知致病变异；Moderate = 同位点另一错义判 Pathogenic；Supporting（第二条）= 旁系同源中 >1 个不同改变判 P/LP
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:21 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

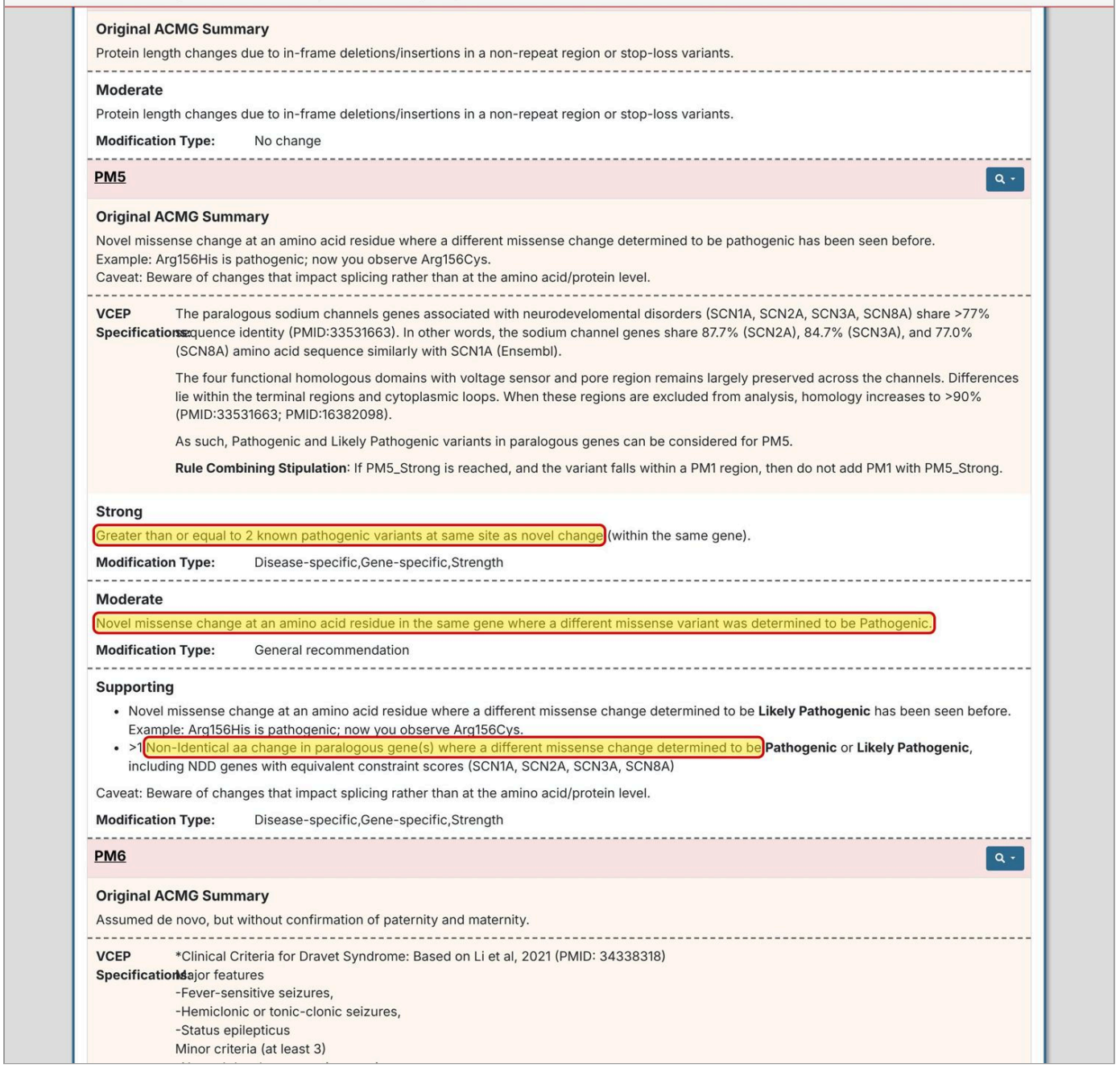


图 D31　VCEP 细则 PM5：Strong = 同位点 ≥2 个已知致病变异；Moderate = 同位点另一错义判 Pathogenic；Supporting（第二条）= 旁系同源中 >1 个不同改变判 P/LP

文件：screenshots/D31_VCEP_PM5.jpg　|　来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067　|　访问：2026-11-05 00:21 UTC

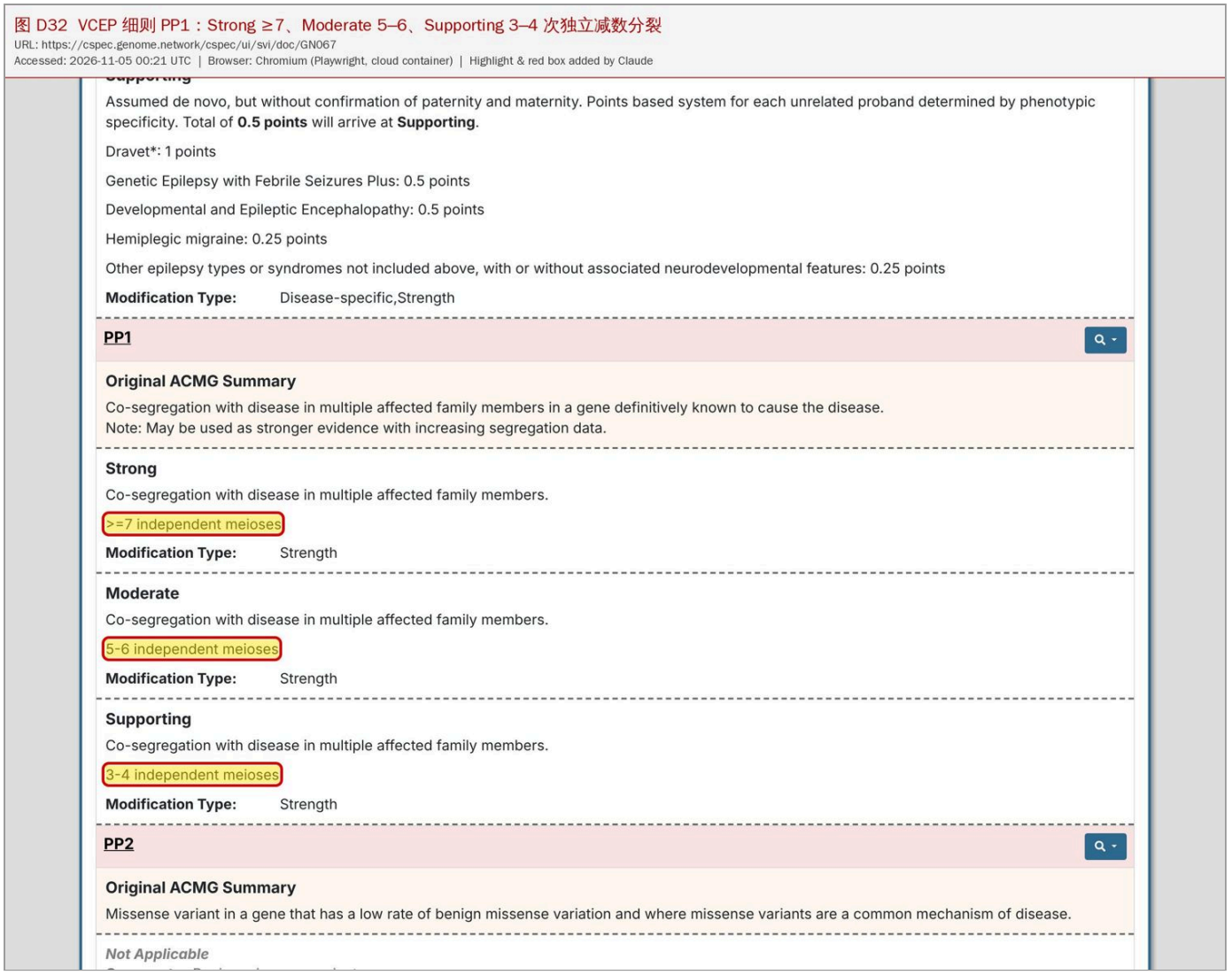


图 D32　VCEP 细则 PP1：Strong ≥7、Moderate 5–6、Supporting 3–4 次独立减数分裂

文件：screenshots/D32_VCEP_PP1.jpg　|　来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067　|　访问：2026-11-05 00:21 UTC

图 D33 VCEP 细则：BA1 = gnomAD AF > 0.02%；BS1 = AF > 0.0004%（均需 ≥5 个等位基因、≥10,000 个已检测）；BS2 Strong = 见于健康成人
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:21 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

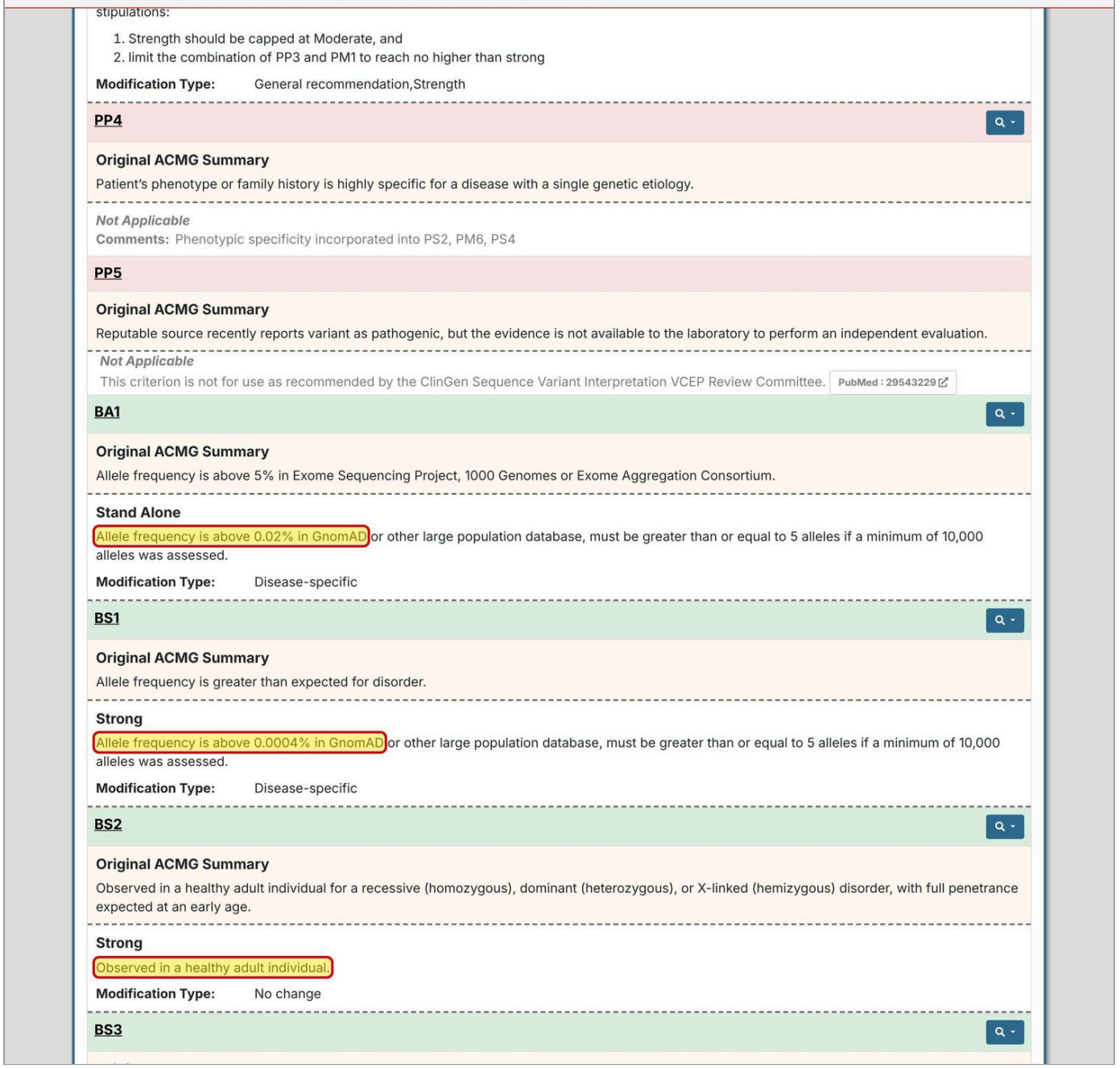


图 D33　VCEP 细则：BA1 = gnomAD AF > 0.02%；BS1 = AF > 0.0004%（均需 ≥5 个等位基因、≥10,000 个已检测）；BS2 Strong = 见于健康成人

文件：screenshots/D33_VCEP_BA1_BS1_BS2.jpg　|　来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067　|　访问：2026-11-05 00:21 UTC

图 D34 VCEP 细则 BP4：Moderate 与 Supporting 均按 ClinGen 推荐（PMID 36413997）用 REVEL 判定
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-05 00:21 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

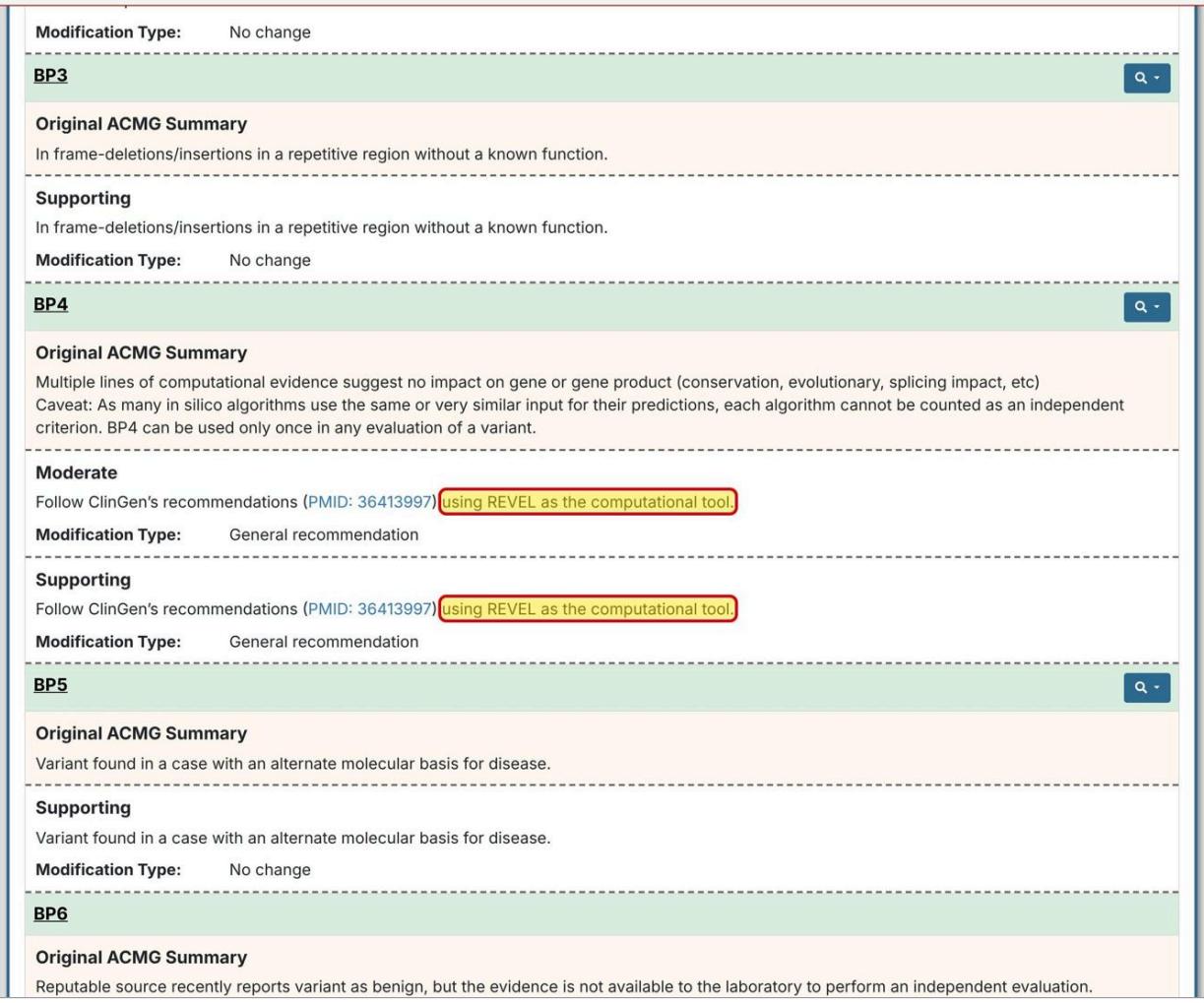


图 D34　VCEP 细则 BP4：Moderate 与 Supporting 均按 ClinGen 推荐（PMID 36413997）用 REVEL 判定

文件：screenshots/D34_VCEP_BP4.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-05 00:21 UTC

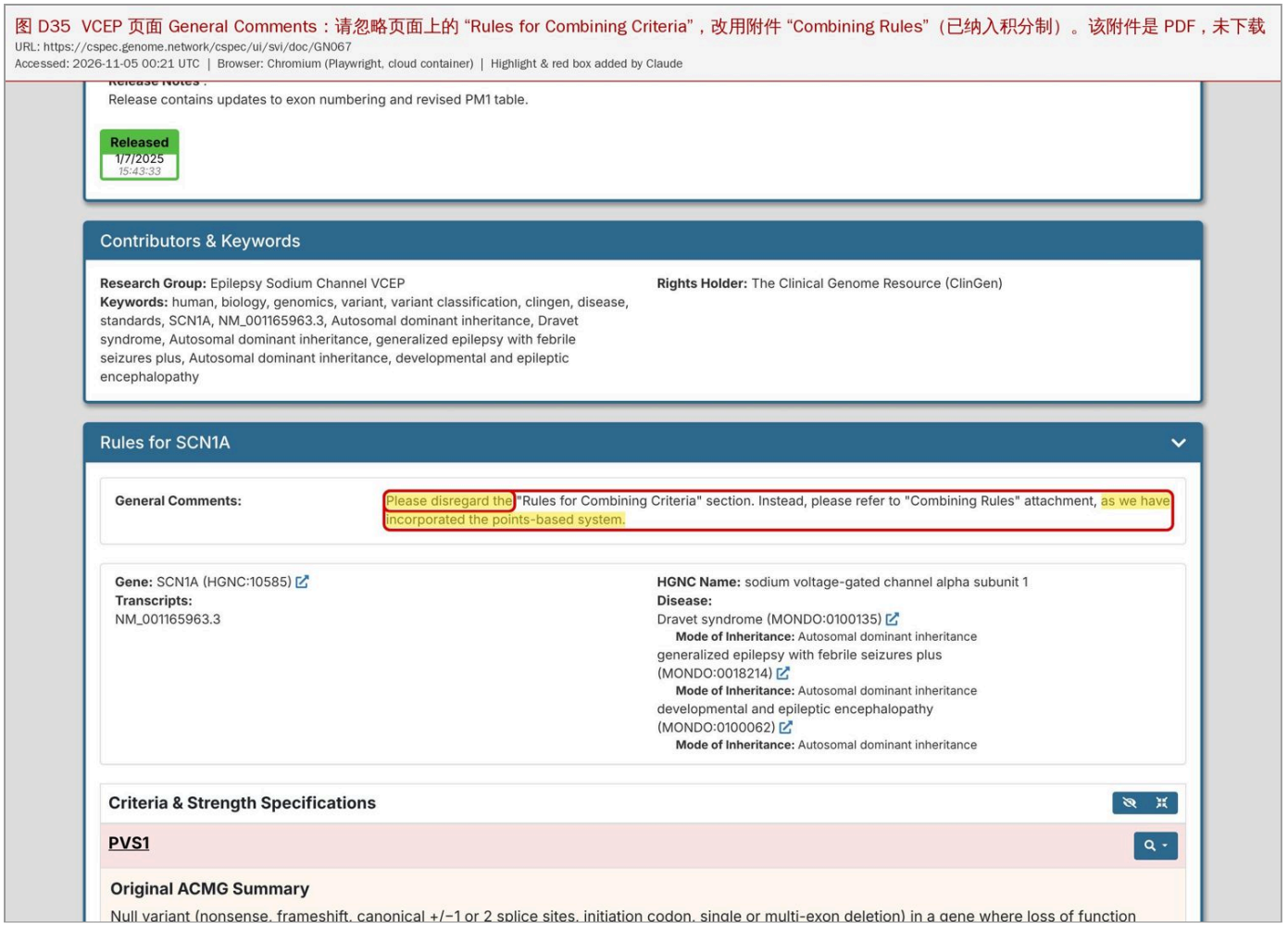


图 D35　VCEP 页面 General Comments：请忽略页面上的 “Rules for Combining Criteria”，改用附件 “Combining Rules”（已纳入积分制）。该附件是 PDF，未下载

文件：screenshots/D35_VCEP_合并规则说明.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-05 00:21 UTC

附录：只用接口或脚本核对、没有截图的项目

- NCBI E-utilities 下载 5 个 SCN1A RefSeq GenBank 记录并翻译（A5 补充说明）；另核对 NM_001165963.3 与 .4 都编码 NP_001159435.1、序列相同（VCEP 细则用的是 .3）。
- PMC 开放获取全文接口：Richards 2015 表 5（ACMG 合并规则）、Tavtigian 2020 表 2/3（积分值与分类区间）。
- UniProt：斑马鱼 scn1laa 全部 5 个条目、旁系同源转录本（经 NCBI GenBank）在对应位点的残基。
- VariantValidator REST：c.3993A>C 同样给出 p.(Glu1331Asp)，GRCh38 g.166009728T>G（网页版每个匿名用户提交次数有限，只截了 C>T 和 A>T）。
- gnomAD GraphQL API：区域内全部变异（含 AC0、AS_VQSR 等过滤变异）与逐位点覆盖度。
- ClinVar E-utilities：SCN1A 全部 5,556 条、SCN2A 3,203 条、SCN3A 2,272 条、SCN8A 2,664 条记录的标题筛选。
- PMC 开放获取全文接口：Pejaver 2022 表 2 的 REVEL 阈值。

待你在自己的 Chrome 补截的项目【需查】

1. NCBI RefSeq NM_001165963.4 页面（MANE Select 关键词、CDS 479..6508、ORIGIN 第 4441 行 “… ctcgatttga agggatgagg”）。

2. Pejaver 2022 表 2（PMC9748256）REVEL 行。
3. 如果报告里要放 ClinVar 网页截图（而不是 E-utilities 输出），在 ClinVar 打开 VCV002036225（SCN2A E1321Q）。
4. UCSC 保守性（可选；已有手动比对代替）。
5. Richards 2015 表 5、Tavtigian 2020 表 2/3 的网页截图（数值已经 PMC 全文接口核对）。

修订记录（2026-11-05 自查）

问题	处理
旧 D21：截图时页面上 REVEL 尚未加载（显示 not available），我的标签却写了 “REVEL 0.613 → 0 points”	作废；等 REVEL 加载完成后重拍，不加标签
旧 D19：标签盖住了 REVEL 分数本身，还写了页面上没有的 “indeterminate”	重拍，不加标签；解释移到图注
D01、D07、D08、D12、D13、D14、D23 的标签压住了部分页面文字	全部重拍为无标签版，说明写进图注
D27 只截到 PS3 的 Strong 和 Moderate，正文却引用了斑马鱼 Supporting 一档	加高窗口重拍，三档都在图内
PS1、PM5、PP1、BA1/BS1/BS2、BP4 的规则和“用附件合并规则”的说明只读了接口文本，没有截图	新增 D30–D35
D09 图注写“约 0.95 的个体覆盖 ≥20×”（目测值，写得像数据）	改为只给 API 数值，并注明截图只能目测
D17 图注写“电压感受器”，但 UniProt 截图里没有这个词	删掉；“S4 是电压感受器”改为需文献支持
结论表写“9 个人类 Nav 旁系同源”（应为 SCN1A 加 8 个旁系同源，共 9 个 α 亚基）	已改正
“最可能的真实变异”写得像结论	拆成可证明部分（只有两种单碱基替换能产生 Asp）和推断部分
斑马鱼 scn1laa 只查了 1 个条目，却推出“可以耐受”	补查全部 5 个条目（都是 D，但同一基因、未审核），结论改为“可能”
第一版 PROGRESS.md 写“29 张都看过高亮位置”，实际只亲眼检查了 11 张	补查了其余全部，发现上述 D19、D21 问题；现 35 张的高亮区域均已亲眼检查（部分通过裁剪红框附近区域查看，不是逐像素看整张图）